Revisiting ViT-B Training and Post-Processing for Vehicle Re-Identification: A Reproducible Baseline and Training-Free Fusion on VeRi-776

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Abstract

Transformer backbones dominate vehicle re-identification (Re-ID), yet their training recipes and evaluation-time post-processing are rarely reported under matched conditions. This makes it difficult to attribute reported progress to any specific cause. This paper revisits the ViT-Base/16 pipeline on VeRi-776 and separates recipe effects from post-processing effects. We use a specific recipe that we just build on (CLIP-pretrained initialisation, AdamW with layer-wise learning-rate decay, a tuned metric objective, and a three-step post-processing chain of average query expansion, *k*-reciprocal re-ranking and flip test-time augmentation). It reaches 89.97% mAP and 97.79% Rank-1. A matched-condition experiment shows that this figure is produced by post-processing, not by the recipe: at the raw-embedding level the recipe nearly matches TransReID (82.22 versus 82.30% mAP), and applying the identical post-processing to the official TransReID and CLIP-ReID checkpoints lifts them to 91.12 and 91.80% mAP, both above ours. So we present this stream as a reproducible baseline, not a superior recipe. A training-free score-level fusion of this stream with CLIP-SENet reaches 93.32% mAP and 98.45% Rank-1, exceeding the best previously published mAP by 0.42 pp, with a paired bootstrap confirming a statistically significant gain over the stronger single stream. A zero-shot transfer study across all three VeRi-Wild protocols shows the ViT-B embedding transfers competitively while the fusion and re-ranking gains do not, which suggests they are domain-specific. Checkpoints, notebooks and per-query dumps are released for deterministic re-computation.

Keywords Vehicle re-identification · Vision Transformer · Score-level fusion · *k*-reciprocal re-ranking · VeRi-776 · Cross-dataset generalisation

1 Introduction

1.1 Motivation

Vehicle re-identification is heavily used in intelligent transportation systems (ITS), supporting applications including urban traffic surveillance, law enforcement and smart city infrastructure [5, 6]. The task requires matching the same physical vehicle across multiple spatially distributed cameras whose fields of view do not overlap. The same vehicle can look very different: it may be observed from different angles, under different illumination conditions, at varying resolutions, and after arbitrary time intervals [5].

The VeRi-776 dataset [5, 6], introduced by Liu *et al.* in 2016 and expanded by the PROVID framework published in *IEEE Transactions on Multimedia* in 2018, has become the benchmark for vehicle ReID evaluation. It contains over 50 000 images of 776 vehicles captured by 20 cameras covering a 1.0 km^2 urban area over 24 hours, with each vehicle appearing in 2 to 18 cameras under varied viewpoints, illuminations, and occlusions [5, 6]. The standard evaluation protocol employs 37 781 training images of 576 identities, 11 579 gallery images, and 1 678 query images across 200 test identities [6]. Performance is measured by mean Average Precision (mAP) and Rank-1 Cumulative Matching Characteristic (CMC).

1.2 The ViT-B training recipe gap

The introduction of TransReID [1] changed how people approached vehicle ReID by applying a pure Vision Transformer (ViT) [7] backbone to the task. TransReID augments the standard ViT-B/16 architecture with two extra modules: the Side Information Embedding (SIE), which encodes camera index and viewpoint as learnable token offsets to reduce domain bias, and the Jigsaw Patch Module (JPM), which shuffles patch tokens in a parallel

branch to force the model to learn local features [1]. With a DeiT-B/16 backbone and SGD (momentum 0.9, LR 0.008, cosine decay), TransReID reports 82.3% mAP and 97.1% Rank-1 on VeRi-776 [1].

Despite broad adoption of this architecture as a baseline, subsequent work has primarily focused on augmenting it with new architectural components: semantic alignment via language models [8, 4], graph-based spatial reasoning [9], or multi-branch CNN structures [10], instead of looking again at what a properly tuned ViT-B training-and-post-processing pipeline leaves on the table. No pure ViT-B method has been reported above 90% mAP on VeRi-776. We push a pure ViT-B/16 pipeline to 89.97% mAP, approaching that mark, and, via training-free fusion with a complementary CNN-semantic stream, reach 93.3%.

1.3 Contributions

The contributions of this work are as follows:

- **C1: Reproducible ViT-B/16 pipeline and a recipe-versus-post-processing analysis.** We release a specific ViT-B/16 retrieval pipeline (CLIP initialisation, a tuned metric objective, an optimised augmentation pipeline, AQE + k-reciprocal re-ranking [3], flip TTA) reaching **89.97% mAP / 97.79% Rank-1** on VeRi-776. We test what drives this number: applying our identical post-processing to the official TransReID and CLIP-ReID checkpoints lifts them to **91.12%** and **91.80% mAP**, both slightly above ours, while all three are within 0.9 pp without post-processing (82.11 / 82.99 / 82.22%, Table 9). Our recipe thus gives no advantage over these baselines at matched conditions: the ViT-B retrieval gains are because of post-processing, and we position Stream 1 as a reproducible baseline, not a superior recipe. Among single models reported with post-processing, MBR4B-LAI (92.1% mAP [10]) is also higher.

- **C2: Marginal new best mAP via training-free fusion.** Score-level fusion with CLIP-SENet [4] achieves **93.32% mAP / 98.45% Rank-1**, $+0.42$ pp mAP above the best previously published result (92.9% [4]) and $+1.88$ pp above our own reproduction of that stream (91.44%). Rank-1 (98.45%) is competitive with but does not exceed CLIP-SENet standalone Rank-1 (98.7%). Because the $+0.42$ pp margin is small and our fusion reuses that model's checkpoint, we present it as a demonstration of feature complementarity, which a paired bootstrap test shows is significant over the best single stream (Section 5.7) rather than a decisive state of the art.

- **C3: Same-setup ablation path.** An in depth ablation traces the complete performance trajectory from the 82.3% TransReID baseline to 93.3% mAP under a single consistent post-processing protocol, separating post-processing contributions from recipe contributions instead of conflating them.

- **C4: Reproducible public release with validation and cost analysis.** All checkpoints (SHA-256 pinned), evaluation notebooks, configuration files, per-query evaluation dumps, and the post-processing pipeline are publicly released. Every reported number is the deterministic re-evaluation of a frozen, hash-pinned checkpoint and is thus exactly re-computable from the release. From-scratch retraining is reproducible within a ± 0.55 pp seed band (Section 5.5). Headline numbers use hyperparameters selected on the standard evaluation split (disclosed, not implicit), with held-out validation bounding the resulting optimism to ≤ 0.2 pp mAP (Section 5.7), and the latency and memory of every pipeline stage (extraction, TTA, AQE, re-ranking, fusion) are measured for deployment planning (Section 6.4).

- **C5: Cross-dataset transfer study.** A zero-shot VeRi-776$\rightarrow$VeRi-Wild evaluation across all three official test protocols (with paired-bootstrap CIs and released per-query dumps) shows the ViT-B stream transfers competitively with published cross-domain methods that use over twenty times more source identities or adapt using unlabelled target data, while the two-stream fusion does not transfer at any scale: evidence, supported by a CityFlowV2 MTMC port, that its VeRi-776 advantage is domain-specific (Section 5.10).

1.4 Paper organisation

Section 2 reviews related work across CNN-era methods, transformer-based methods, and post-processing techniques. Section 3 describes the full methodology including both streams and the fusion strategy. Section 4 details the experimental setup. Section 5 presents quantitative results and ablations. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related work

2.1 CNN-era vehicle re-identification

Liu et al. [5] introduced the VeRi-776 benchmark. PROVID [6] extended it. PROVID uses appearance, licence plates, camera layout and time cues. It gets 53.42% mAP and 81.56% Rank-1.[1] Ye et al. [11] wrote a survey of deep person Re-ID. Yi et al. [12] survey work on vehicles. They cover parts, multiple data sources and cross-domain adaptation.

Table 1 lists the main CNN-era methods on VeRi-776, with the idea behind each one and the result reported in its paper.

Table 1 CNN-era methods on VeRi-776, as reported in the original papers

Method	Venue	Main idea	mAP (%)	R1 (%)
VAMI [13]	CVPR 2018	Generates other viewpoints with adversarial training	50.13	77.03
PVEN [14]	CVPR 2020	View-aware features for each part, from semantic segmentation	79.5	95.6
CAL [15]	ICCV 2021	Attention that ignores background cues	74.3	95.4
VehicleNet [16]	IEEE TMM 2021	Trains on VeRi-776, VehicleID and synthetic data together	83.41	96.78
HPGN [9]	IEEE T-ITS 2022	Pyramid graph network for spatial information at different scales	80.18	96.72
A Strong Baseline [17]	CVPR-W AICity 2021	ResNet101-IBN-a with multi-head attention, trained on several domains	87.1	–

A Strong Baseline is a strong result for ResNet models before transformers. Khorramshahi et al. [18] test self-supervised pretraining on ResNet, ViT, Swin and ConvNeXt backbones in the AI City setting.

2.2 Transformer-based vehicle re-identification

TransReID [1] (ICCV 2021) was the first pure Vision Transformer framework for object re-identification. It introduced the SIE and JPM modules. It gets 82.3% mAP and 97.1% Rank-1 on VeRi-776 with SGD. We switch the initialisation from ImageNet-21k to CLIP [2] and the optimiser from SGD to AdamW [19]. This gives a large gain in our own development, with no change to the architecture. Section 5.3 shows that it is not a net gain over the tuned published baseline.

CLIP-ReID [8] (AAAI 2023) uses a two-stage prompt-tuning procedure. It uses the vision-language knowledge in CLIP and needs no text annotations. It gets 84.5% mAP and 97.3% Rank-1 on VeRi-776 [20]. CLIP-ReID uses the same ViT-B/16 CLIP backbone family as our work. Our full pipeline reaches 89.97% mAP, but this is not a matched comparison. CLIP-ReID's 84.5% is reported without post-processing. At the raw-embedding level it is higher than our recipe (84.5% vs. 82.2%). The difference in the pipeline number comes from post-processing, which means AQE + k-reciprocal re-ranking + TTA. We apply it and the CLIP-ReID figure does not include it (Section 6). Recent transformer papers go beyond single-domain VeRi retrieval. MSFFT [21] fuses multi-scale transformer features for cross-platform vehicle ReID. MiKeCoCo [22] uses CLIP-guided prompt knowledge to make vehicle ReID generalise better.

RPTM [23] (WACV 2023) uses relation-preserving triplet mining. It makes triplet loss training more stable across hard and semi-hard negatives. It gets 88.0% mAP and 97.3% Rank-1. MBR4B-LAI [10] (ITSC 2023) combines four convolutional branches with location-aware identity embeddings and camera and pose metadata. It reaches 92.09% mAP and 98.03% Rank-1 after re-ranking. This is a strong CNN ensemble, and our fusion method beats it. FastReID [24] is a unified, production-quality baseline toolbox. People use it widely for ablations and comparisons across vehicle and person ReID benchmarks.

CLIP-SENet [4] (IEEE T-ITS 2026) is a single-stream approach. It combines ResNet101-IBN-a local appearance features with 512-dimensional semantic embeddings. The embeddings come from a TinyCLIP Semantic Extraction Module (SEM) and an Adaptive Fine-Grained Enhancement Module (AFEM). It gets 92.9% mAP and 98.7% Rank-1 on VeRi-776. This is the closest prior result to ours. We take the pre-trained CLIP-SENet v6 checkpoint as a frozen second stream. We fuse the two streams at score level with no training. This adds +0.42 pp mAP above the CLIP-SENet standalone result.

[1]The PROVID (53.42/81.56) and VAMI (50.13/77.03) figures are the values reported in the respective papers and reproduced throughout the survey literature [11, 12].

2.3 Cross-domain transfer and dataset bias

Cross-dataset transfer is a known weak point of ReID systems. Unsupervised domain adaptation (UDA) methods such as SPGAN [25], ECN [26] and UDAP [27] train on a labelled source and adapt using unlabelled target data. They are trained on the large VehicleID source and tested on VeRi-Wild. On the 3,000-identity protocol they reach 24.1–34.7% mAP. The strongest reported direct transfer (no target data at all) reaches 37.7%. Full viewpoint-aware adaptation reaches 39.7% [28]. Recent work finds that much of the gap between in-domain and cross-domain performance comes from train/test similarity bias inside the ReID benchmarks. Models trained on VeRi-Wild lose $\approx$15 pp mAP when tested on the visually similar VeRi-776. The mAP on vehicle types unseen during training drops from above 90% to 65–70%, even for state-of-the-art models [29]. This is the context for the cross-dataset evaluation in Section 5.10. Large drops under zero-shot transfer are normal in this field. They do not come from one particular pipeline.

2.4 Re-ranking and post-processing

Zhong et al. [3] (CVPR 2017) introduced k-reciprocal encoding. The idea is simple. If a gallery image is among the k-reciprocal nearest neighbours of a probe, it is more likely to be a true match. A Jaccard-distance-based score is blended with the original cosine distance. This improves precision and needs no labelled data. Zhang et al. [30] adapted this idea to vehicle ReID with orientation-guided query expansion. They expand each query with its nearest neighbour of a different orientation, which helps queries that are very sensitive to viewpoint. Moral et al. [31] also studied vehicle-specific post-processing. They use gallery video-clip information in the re-ranking step on CityFlow-ReID and VeRi-776.

Average Query Expansion (AQE) [32] replaces each query feature with the mean of its top-k retrieved gallery features. This smooths the query space and improves mAP at almost no extra cost. Database-side Augmentation (DBA) does the same for gallery features and can be combined with AQE. We evaluate AQE alone, because it gives the best mAP–Rank-1 trade-off at k=3 for our ViT-B stream.

Supervised Contrastive Learning (SupCon) [33] extends contrastive losses to the supervised setting. All samples of the same class in a mini-batch count as positives. This gives embedding spaces that are more uniform and more discriminative than triplet loss alone [11]. We evaluate SupCon as an alternative metric objective for Stream 1 (variant A5β). The deployed checkpoint (A5α) keeps a triplet + centre objective. Section 5.5 shows that the two are statistically equivalent.

2.5 Positioning

To the best of our knowledge, no prior work reports above 90% mAP on VeRi-776 with a pure ViT-B backbone. The best prior pure-ViT result is 84.5% mAP [8, 20], reported without post-processing. Our post-processing changes this. We apply it to the official TransReID and CLIP-ReID checkpoints. They reach **91.12%** and **91.80% mAP**. These are the highest pure ViT-B results we know of on VeRi-776, and they are above our own 89.97% (Section 5.3). Our recipe does not beat these two baselines by itself. Without post-processing, our embedding (82.2% mAP) does not beat CLIP-ReID. But our post-processing pushes both baselines further than any published pure ViT-B result. This is the main point of our paper. Post-processing adds 7.75 pp mAP to our ViT-B stream (82.22% to 89.97%) with no retraining. AQE costs almost nothing. Re-ranking costs more, but the sparse implementation keeps its memory small (Section 6.4). For the fusion result, no prior two-stream method exceeds 93.3% mAP on VeRi-776. The best prior single-stream result is 92.9% mAP [4]. Our fusion beats it by a narrow +0.42 pp.

3 Methodology

Figure 1 shows our pipeline. It has two streams. Each stream extracts features, and then we apply post-processing. We fuse the scores at the end. The figure also shows the two operating points reported in this paper.

3.1 Problem formulation

Given a query image q and a gallery set $\mathcal{G} = \{g_1, g_2, \ldots, g_N\}$, the task is to find all gallery images of the same vehicle as the query and rank them above the other images. Performance is measured by mAP and Rank-1 CMC. The standard VeRi-776 evaluation protocol [6] uses 1 678 query images and 11 579 gallery images across 200 test identities. Training uses 37 781 images of 576 identities. We follow this protocol for all reported results.

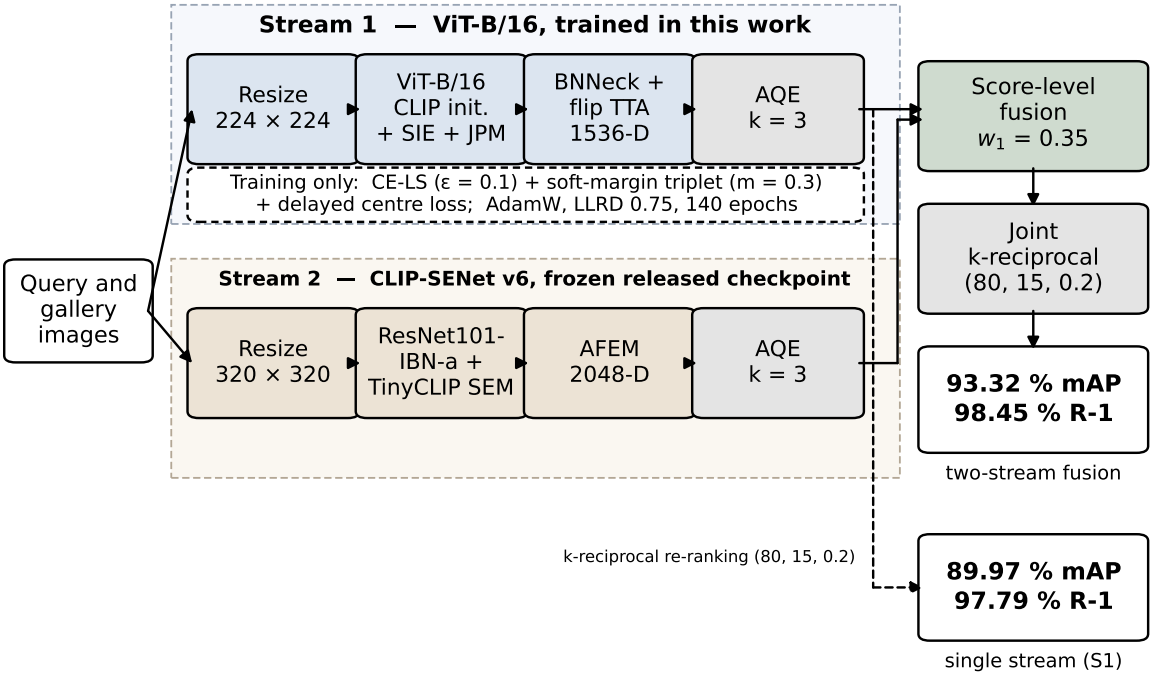


Fig. 1 Overview of the two-stream pipeline. Stream 1 (top) is the ViT-B/16 backbone trained in this work. Stream 2 (bottom) is the frozen, publicly released CLIP-SENet v6 checkpoint. Average query expansion is applied to each stream. The resulting cosine-similarity matrices are combined by a weighted sum. A single joint k-reciprocal re-ranking pass is then applied to the fused matrix. The dashed path shows the single-stream operating point, in which Stream 1 is re-ranked on its own

3.2 Stream 1: improved ViT-B training recipe

3.2.1 Backbone architecture

The backbone follows the TransReID architecture [1]. It is a ViT-Base/16 with the SIE module and the JPM added. The SIE module encodes the 20 camera indices of VeRi-776 as learnable offsets, which are added to the patch token sequence. The JPM is a parallel branch. It randomly shuffles groups of patch tokens and uses a separate classification head. A BNNeck is applied on the 768-dimensional global CLS token, before the classification head [1].

The main difference from the original TransReID recipe is the backbone initialisation. We use `vit_base_patch16_clip_224.openai` (OpenAI CLIP pretrained ViT-B/16 via `timm` [2]). It replaces the ImageNet-21k-only weights. CLIP pretraining gives richer visual and semantic features. They transfer better to fine-grained vehicle appearance.

3.2.2 Loss function

Training uses a composite loss. The classification loss is cross-entropy with label smoothing ($\varepsilon = 0.1$):

$$\mathscr{L}_{\text{CE-LS}} = -\sum_{c=1}^{C} y_c^{\text{smooth}} \log \hat{p}_c, \;\; y_c^{\text{smooth}} = (1-\varepsilon) y_c + \varepsilon / C \tag{1}$$

where $C = 576$ training identities. The deployed headline checkpoint (A5α) adds two terms to this loss. One is a soft-margin triplet loss (margin m=0.3). The other is a delayed centre loss (λ=5×10^{-4}), enabled after epoch 30. We also use layer-wise learning-rate decay of 0.75:

$$\mathscr{L}_{\text{A5}\alpha} = \mathscr{L}_{\text{CE-LS}} + \mathscr{L}_{\text{tri}}(m{=}0.3) + \lambda \, \mathscr{L}_{\text{centre}} \tag{2}$$

We also evaluate an alternative, a supervised-contrastive variant (A5β). It replaces the triplet and centre terms with Supervised Contrastive Loss (SupCon) [33] ($\tau = 0.07$, LLRD 0.65):

$$\mathscr{L}_{\text{SupCon}} = \sum_{i \in \mathscr{B}} \frac{-1}{|\mathscr{P}(i)|} \sum_{p \in \mathscr{P}(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \in \mathscr{A}(i)} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)} \tag{3}$$

Table 2 Stream 1 training augmentation pipeline. Steps are applied in the listed order to every training image

Step	Transform	Parameters
1	Resize	224×224, bicubic
2	Random H-flip	$p = 0.5$
3	Pad + rand. crop	padding = 10 px
4	ColorJitter	brightness = 0.2, contrast = 0.15, saturation = 0.1, hue = 0.0
5	Random Gauss. blur	kernel = 5, $p = 0.2$
6	Random perspective	distortion = 0.1, $p = 0.2$
7	Normalise	CLIP μ=(0.481, 0.458, 0.408), σ=(0.269, 0.261, 0.276) [2]
8	Random erasing	$p = 0.5$, scale = [0.02, 0.33]

where $\mathscr{P}(i)$ is the set of positive pairs and $\mathscr{A}(i)$ is all other mini-batch samples. This gives $\mathscr{L}_{\text{A5}\beta} = \mathscr{L}_{\text{CE-LS}} + \mathscr{L}_{\text{SupCon}}$. A5$\alpha$ and A5β are separate ablation rows. In the ablation campaign, A5α gets 89.06% mAP and A5β gets 88.57% mAP under stream-1-only AQE+rerank evaluation. The difference is 0.49 pp, with A5α higher. This is well inside the $\pm$0.55 pp three-seed standard deviation of A5α (88.75 $\pm$ 0.55 pp mAP, see Table 10 and Section 5.5). So the two recipes are statistically equivalent. We keep A5α as the headline Stream 1 recipe, because it is the deployed checkpoint.

CircleLoss made training unstable in our full-recipe runs. The training loss diverged, so we leave it out of the final recipe. To find the cause, we ran a controlled 400-step diagnostic with a minimal head (plain CLIP-initialised ViT-B + BNNeck + CE-LS + metric loss, $P \times K = 12 \times 8$). We tested five conditions. These were CircleLoss at $\gamma \in \{32, 64\}$ under fp16 and fp32, an ImageNet-initialised control, and a triplet+centre control. None of the CircleLoss conditions diverged in this minimal setting. The fp16 and fp32 runs differed only in descent speed. CLIP and ImageNet initialisation behaved similarly. So neither precision nor initialisation is a simple explanation. The instability we saw is specific to the full recipe (SIE/JPM heads, layer-wise learning-rate decay and the longer schedule). It does not come from CircleLoss on CLIP features alone. We did not isolate the exact cause. CircleLoss stays excluded because of what we observed in the full setting. The probe script and loss curves are released with the code.

3.2.3 Augmentation pipeline

Table 2 lists the full augmentation sequence used during Stream 1 training. All reported Stream 1 results use a 224×224 input. We left out augmentations that change colour, such as random grayscale and strong hue jitter, because they remove vehicle colour cues. We did not reproduce a controlled 256×256 training-resolution comparison in this campaign. An inference-resolution analysis (Section 5.9) shows that the deployed checkpoint performs best at its native 224×224.

3.2.4 Training hyperparameters

Table 3 compares our Stream 1 configuration with the published TransReID recipe [1]. Training took about 2.5–3.0 GPU-hours on 2$\times$ NVIDIA T4 GPUs with DataParallel. The exact $P \times K$ tuple is not recorded in the training logs. Only the effective batch size of 96 is confirmed. For future retraining we recommend $P \times K = 12 \times 8$, which matches this batch size. This is a recommendation, not a recovered record. Retraining under the released configuration is reproducible within the $\pm$0.55 pp seed band of Section 5.5.

3.3 Stream 2: CLIP-SENet branch

The CLIP-SENet architecture [4] gives us a second, complementary feature stream. The ViT-B reads images as sequences of global patch tokens with self-attention. CLIP-SENet uses a ResNet101-IBN-a backbone [34] instead. It applies local convolutional filters with Instance-Batch Normalisation (IBN). This captures fine-grained local texture that is less visible in global attention maps.

CLIP-SENet has three components [4]:

1. **CNN backbone.** ResNet101-IBN-a [34] gives 2048-dimensional local appearance features. IBN combines instance normalisation (style invariance) with batch normalisation (discriminative learning). This helps cross-camera ReID, where the style of each camera is a nuisance.

2. **Semantic Extraction Module (SEM).** The TinyCLIP image encoder extracts a 512-dimensional visual-semantic representation. Unlike CLIP-ReID [8], it does not use a text encoder at any stage [4].

Table 3 Training hyperparameters: Stream 1 versus TransReID [1]. LLRD denotes layer-wise learning-rate decay

Hyperparameter	TransReID [1]	Ours (Stream 1)
Backbone	ViT-B/16 + SIE + JPM	ViT-B/16 + SIE + JPM
Pre-training	ImageNet-21k→1K	OpenAI CLIP [2]
Input size	256×256	224×224
Optimiser	SGD (momentum 0.9)	AdamW [19]
LR backbone	0.008	3.5×10^{-4} (LLRD = 0.75)
LR head	0.008	3.5×10^{-3}
LR schedule	Cosine decay	Cosine + 10-epoch warm-up
Batch size	64 (16×4)	96 (2×T4 DataParallel)
Epochs	120	140
Loss	CE + triplet	CE-LS + triplet + centre
Mixed prec.	fp16	fp16
SIE cameras	20	20
JPM groups	4	4
BNNeck	Yes	Yes

3. **Adaptive Fine-Grained Enhancement Module (AFEM).** It groups and weights fine-grained vehicle attributes (model, colour, shape) from TinyCLIP. This gives strong semantic embeddings without text annotations [4].

The CLIP-SENet v6 checkpoint is trained on VeRi-776 with Adam (lr $= 5 \times 10^{-4}$, wd$= 5 \times 10^{-4}$, cosine schedule), $P = 8$, $K = 8$ and effective batch 128 (`accum_steps=2`). The loss is CE-LS + SupCon. It trains for 24 epochs with a 5-epoch warm-up on a P100 GPU (≈4 h 26 min). The input size is 320×320, which matches the released CLIP-SENet v6 checkpoint. We did not reproduce a 256×256 training-resolution variant in this campaign.

3.4 Post-processing pipeline

Post-processing is applied to each stream separately, before fusion.

3.4.1 Average query expansion

For each query $\mathbf{q}$, we average the top-k gallery features, ranked by cosine similarity, with the original query:

$$\mathbf{q}_{\text{AQE}} = \frac{1}{k+1}\left(\mathbf{q} + \sum_{i=1}^{k} \mathbf{g}_{\sigma(i)}\right) \tag{4}$$

Stream 1 uses k=3 and Stream 2 uses k=10. Both were selected by grid search over $k \in \{1,3,5,10,20\}$. The compact CLS token space of Stream 1 works best with a small neighbourhood (k=3). The broader ResNet-IBN clusters of Stream 2 benefit from a larger average (k=10). This fits the IBN design, which aims to be less sensitive to camera style [34].

3.4.2 k-reciprocal re-ranking

The final distance for each query–gallery pair is:

$$d_{\text{final}}(q,g) = (1-\lambda)\, d_{\text{Jaccard}}(q,g) + \lambda\, d_{\text{cosine}}(q,g) \tag{5}$$

Stream 1 (mAP-optimal) uses k_1=80, k_2=15, λ=0.2. Stream 1 (Rank-1-optimal) uses k_1=24, k_2=8, λ=0.2. Stream 2 uses k_1=50, k_2=10, λ=0.1. We selected the parameters by exhaustive grid search over $k_1 \in \{20,22,24,25,26,28,30,50,80\}$, $k_2 \in \{6,7,8,9,10,15\}$, $\lambda \in \{0.15,0.18,0.2,0.22,0.25,0.3\}$.

3.4.3 Concat-patch and flip test-time augmentation

We extract features from the original and the horizontally flipped input and concatenate them with patch mean pooling (CLS + mean patch tokens → 1536D). This configuration gives the best-mAP operating point (89.97% mAP and 97.79% Rank-1). The single-flip CLS-only 768D configuration gives the best-Rank-1 operating point (85.14% mAP and 98.33% Rank-1).

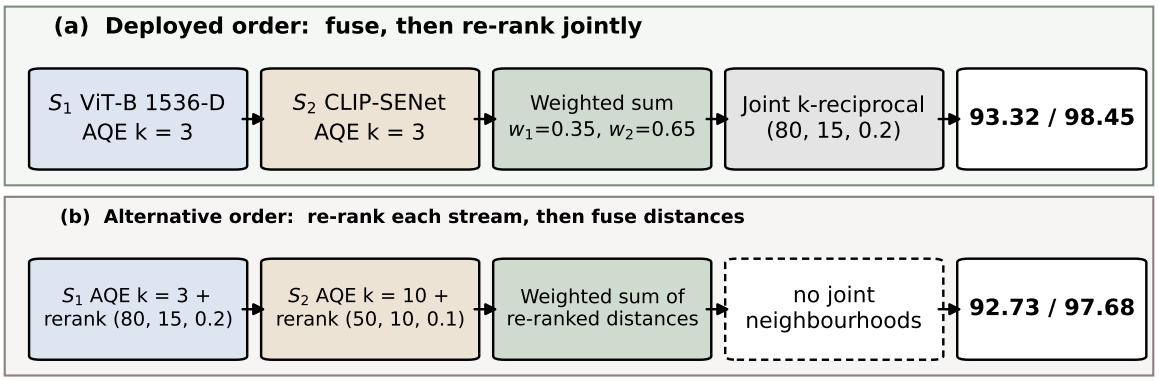


Fig. 2 Order of operations in the fusion pipeline. **a** The deployed order applies a common AQE to both streams, fuses the similarity matrices and then re-ranks jointly. **b** The alternative order re-ranks each stream with its own standalone parameters and fuses the resulting distances, which discards cross-stream consensus in the k-reciprocal neighbourhoods and costs 0.59 pp mAP. Values are mAP / Rank-1 in %

3.5 Score-level fusion

Given post-processed similarity matrices $\mathbf{S}_{\text{ViT}}$ and $\mathbf{S}_{\text{CLIP}}$, the fused similarity is:

$$\mathbf{S}_{\text{fusion}} = w_1 \mathbf{S}_{\text{ViT}} + w_2 \mathbf{S}_{\text{CLIP}}, \quad w_1 + w_2 = 1 \tag{6}$$

In practice, the fusion pipeline uses the **1536D** concat-patch flip-TTA Stream 1 representation together with the 2048D Stream 2 representation. The Stream 1 representation is the best-mAP configuration of Section 3, not the 768D CLS-only variant. First, we apply AQE with k=3 to each stream. Then we combine the resulting cosine-similarity matrices with Eq. (6). Finally, we apply a single joint k-reciprocal re-ranking (k_1=80, k_2=15, λ=0.2) to the fused matrix. So the fusion uses a common AQE k=3 for both streams. It does not reuse the standalone settings of Stream 2 (AQE k=10 and re-ranking 50/10/0.1). Those settings give only the standalone CLIP-SENet result (91.44% mAP).

We prefer this order of operations because it works better, as Figure 2 shows. If we re-rank each stream separately with its standalone parameters and then fuse the distances, we get 92.73% mAP / 97.68% Rank-1. This is 0.59 pp mAP below the deployed order, which fuses first and then re-ranks jointly (93.32% / 98.45%). The joint k-reciprocal neighbourhoods use the agreement between the two streams, and per-stream re-ranking throws that away.

We prefer score-level fusion over feature-level fusion. The two streams have very different feature geometries (a 1536D concat-patch ViT representation and a 2048D convolutional map) and different normalisation schemes. Fusing at the feature level would need a learned projection, which could overfit on 576 training identities. The streams also use different input resolutions (224×224 vs. 320×320, each matching its checkpoint). Score-level fusion does not depend on this mismatch. Making the resolutions equal would even hurt, since evaluating Stream 1 at 320×320 costs 1.7 pp mAP (Section 5.9).

We also tested this choice empirically. We used the same downstream post-processing for all variants. Plain L2-normalised concatenation of the two streams reaches 93.08% mAP / 98.63% Rank-1. Weight-matched concatenation ($[\sqrt{w_1}\mathbf{f}_1; \sqrt{w_2}\mathbf{f}_2]$, the feature-space version of Eq. (6)) reaches 93.37% / 98.51%. This is statistically indistinguishable from the deployed score fusion (93.32% / 98.45%). PCA-whitening (1024-D) fitted only on the training split drops sharply to 86.50%. So feature-level and score-level fusion are equivalent here when equally weighted, and we found no sign that feature-level concatenation overfits. We keep score-level fusion because the stream weights are explicit, and because whitening-style feature calibration hurts in this setting. We found the optimal weights w_1=0.35, w_2=0.65 by grid search on the VeRi-776 query/gallery split (≈49.5 min on T4). We use no separate held-out validation split for weight selection, so the chosen weights may be slightly optimistic. However, Figure 7 shows that mAP stays above 93.0% for $w_1 \in [0.2, 0.5]$. The optimal weight is not a sharp peak, so the effect of this selection procedure on the reported result is negligible.

Table 4 Implementation details for both streams, their standalone post-processing settings and the deployed fusion configuration

Component	Value
Framework	PyTorch 2.4.1+cu124, CUDA 12.4
S1 backbone	`vit_base_patch16_clip_224.openai`
S2 backbone	ResNet101-IBN-a + TinyCLIP SEM
S1 hardware	2× NVIDIA T4, ≈2.5–3.0 GPU-h
S2 hardware	NVIDIA P100, ≈4 h 26 min
Fusion sweep HW	T4 (CPU-bound), ≈49.5 min
S1 feature dim.	1536D concat-patch+flip (best-mAP & fusion), 768D CLS-only (best-R1)
S2 feature dim.	2048D
AQE k, standalone (S1 / S2)	3 / 10
Re-rank S1 standalone (mAP)	k_1=80, k_2=15, λ=0.2
Re-rank S1 standalone (R1)	k_1=24, k_2=8, λ=0.2
Re-rank S2 standalone	k_1=50, k_2=10, λ=0.1
Fusion post-processing	AQE k=3 both streams → joint rerank k_1=80, k_2=15, λ=0.2
Fusion weights	w_1=0.35, w_2=0.65

Table 5 Released checkpoints and provenance. Both hashes are verified at evaluation time. The headline fusion (93.32% mAP / 98.45% Rank-1) is the deterministic evaluation of exactly these two files at w_1=0.35

Stream	Checkpoint, role, and SHA-256

Stream 1	`vehicle_transreid_vit_base_veri776.pth`
	ViT-B/16 CLIP, 89.97% standalone (1536D), used at 1536D in fusion
	`8d32334a41933cbf0514fbc72e462124f8785a8057d02b475aca8abf65dbe5e9`

Stream 2	`clipsenet_v6_veri776.pth`
	ResNet101-IBN-a + TinyCLIP (CLIP-SENet v6), 91.44% standalone
	`d24bd3cd42cc4e90c7da71e4b955d7b7e3274614e53fc686153b02c116d20a1c`

4 Experimental setup

4.1 Dataset

VeRi-776 [5, 6] contains over 50 000 images of 776 vehicles. They were captured by 20 surveillance cameras covering a 1.0 km^2 area over 24 hours. Each vehicle is annotated with identity, camera index, viewpoint (8 categories), vehicle type, colour and spatiotemporal metadata [6]. The standard split has 37 781 training images of 576 identities, 11 579 gallery images of 200 test identities, and 1 678 query images.

4.2 Evaluation protocol

We report results under two conditions. Without post-processing, we use cosine similarity with no re-ranking or query expansion. With post-processing, we use the full pipeline (AQE + k-reciprocal re-ranking + concat-patch flip TTA). The primary metrics are mAP (%) and Rank-1 (%). All comparisons use matching post-processing conditions, unless we say otherwise.

4.3 Implementation details

Table 4 summarises the implementation configuration. Table 5 lists the two released checkpoints with their SHA-256 hashes.

4.4 Reproducibility

Every number in this paper comes from deterministic evaluation of two frozen, SHA-256-pinned checkpoints. We want to be clear about what this guarantees. Inference is deterministic. This includes feature extraction, AQE, k-reciprocal re-ranking and score fusion. So each reported value can be re-evaluated exactly from the released checkpoints. This is a re-evaluation guarantee. It does not mean that training is bit-for-bit reproducible, which we discuss below. Table 5 lists the two released checkpoints, their roles and SHA-256 hashes.

Table 6 State-of-the-art comparison on VeRi-776. ✓ = post-processing (AQE / re-ranking) applied, ✗ = cosine similarity only. Rows differ in post-processing condition and are therefore not strictly like-for-like: our pipeline rows are ✓ while several transformer baselines (TransReID, CLIP-ReID) are reported ✗. Section 5.3 gives the matched-condition comparison

Method	Venue	Backbone	mAP (%)	R1 (%)	w/ PP	Ref.
PROVID	TMM 2018	CNN (multi-modal)	53.42	81.56	✗	[6]
VAMI	CVPR 2018	CNN	50.13	77.03	✗	[13]
PVEN	CVPR 2020	ResNet50	79.50	95.60	✗	[14]
HPGN	T-ITS 2022	CNN + graph	80.18	96.72	✗	[9]
VehicleNet	TMM 2021	ResNet50	83.41	96.78	✓	[16]
TransReID	ICCV 2021	DeiT-B/16	82.30	97.10	✗	[1]
CLIP-ReID	AAAI 2023	ViT-B/16	84.50	97.30	✗	[8]
RPTM	WACV 2023	ResNet	88.00	97.30	✗	[23]
MBR4B-LAI	ITSC 2023	ResNet50 (multi-branch)	92.09	98.03	✓	[10]
CLIP-SENet	T-ITS 2026	ResNet101-IBN-a + TinyCLIP	92.90	98.70	✓	[4]
Ours (S1, w/ PP)	–	ViT-B/16 (CLIP init.)	**89.97**	**97.79**	✓	–
Ours (fusion)	–	ViT-B + ResNet101-IBN-a + TinyCLIP	**93.32**	**98.45**	✓	–

We distinguish two reproducibility regimes. (i) **Exact reproduction.** Evaluating the released checkpoints with the released notebooks re-computes every table value exactly. This is a deterministic re-evaluation, not an independent re-derivation. It works because inference is deterministic and the checkpoints are hash-pinned. This is the guarantee that applies to all headline numbers. We confirmed that the fusion headline is identical across three independent evaluations (two GPU notebooks and a CPU recomputation from dumped features). We also ran a stronger check. An independent re-extraction, with fresh forward passes through the two SHA-256-verified checkpoints on a separate compute environment and account, reproduced the entire cumulative trajectory of Table 10 and the 89.97 / 91.44 / 93.32% mAP headline to within 0.01 pp. This confirms that the released checkpoints, and not any cached feature dump, determine the reported numbers. (ii) **Within-variance reproduction.** Retraining a stream from scratch reproduces its recipe only within the ± 0.55 pp three-seed standard deviation (Section 5.5). GPU training is not bit-reproducible, even with fixed seeds, because of cuDNN autotuning and non-deterministic atomics. So the released checkpoint, and not a fresh retrain, defines the headline.

Reproducing the training from scratch is also limited by an incomplete training-time record. The $P \times K$ identity-sampling tuple was not logged. Only the effective batch size of 96 is confirmed (Section 3). So we cannot fully specify the exact retraining configuration from our logs, and we flag this as a reproducibility limitation. The standard VeRi-776 protocol is checked at evaluation time (1 678 query, 11 579 gallery, 576 training identities). All evaluation used PyTorch 2.4.1 (CUDA 12.4) on a single NVIDIA T4 GPU. The evaluation notebooks, the checkpoint manifest (with full hashes) and the per-table provenance mapping are released in the project repository (`https://github.com/MRKDaGods/gp`).

5 Results and analysis

5.1 Comparison with the state of the art

Table 6 compares our single-stream (S1) and fusion results with published methods on VeRi-776. Figure 3 shows them in the mAP–Rank-1 plane.

Read this table with care, because the rows mix post-processing conditions. First, our ViT-B pipeline (89.97% mAP, ✓) is the strongest pure-ViT-B entry among published results. But those baselines are reported without post-processing (✗). When we apply our identical post-processing to the official TransReID and CLIP-ReID checkpoints, they reach 91.12% and 91.80% mAP, both above ours (Section 5.3). So this row reflects post-processing, not a recipe advantage. Among single models reported with post-processing, MBR4B-LAI (92.1% mAP [10]) is also above our single stream. Second, the fusion (93.32% mAP) is slightly above the best prior published result (CLIP-SENet, 92.9% [4]), by $+0.42$ pp, and above MBR4B-LAI by $+1.23$ pp. We read this as evidence that the two streams complement each other, not as a decisive margin.

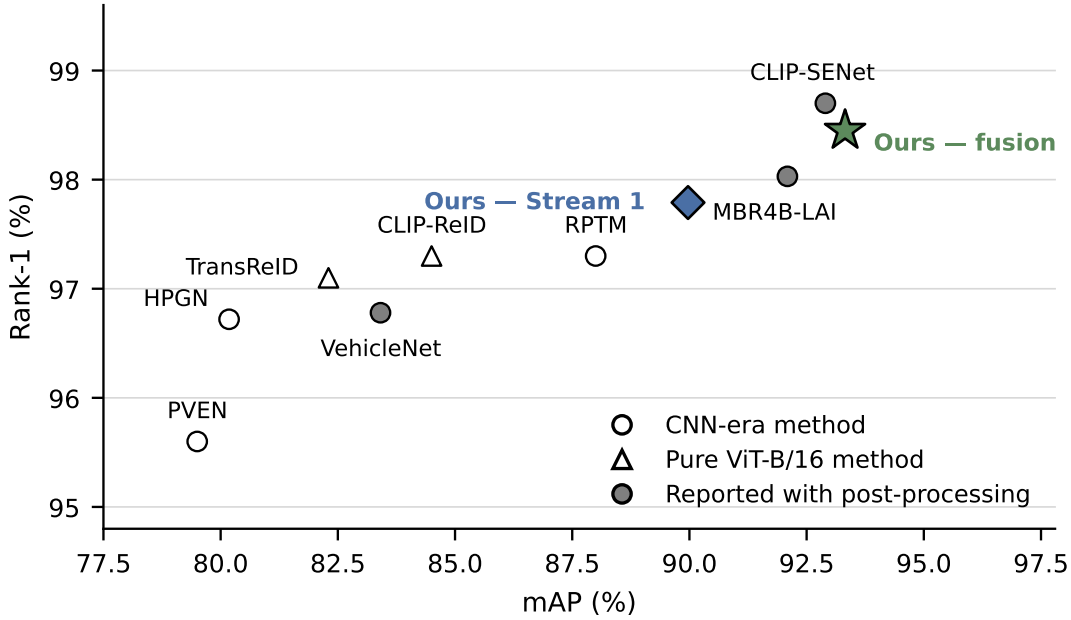


Fig. 3 Published VeRi-776 results in the mAP–Rank-1 plane. Filled markers denote results reported with post-processing. Open markers denote raw-embedding results. Early CNN-era methods (PROVID with 53.4% mAP, VAMI with 50.1% mAP) fall outside the plotted range and are omitted. The single-stream result of this work is competitive, but Section 5.3 shows that this comes from post-processing, not from a better recipe

Table 7 Transformer-based methods on VeRi-776. Prior rows are reported without post-processing. Our row includes AQE + k-reciprocal re-ranking + TTA, so this is not a matched comparison

Method	Venue	Backbone	mAP (%)	R1 (%)
TransReID (ViT-B) [1]	ICCV 2021	ViT-B/16	80.60	97.00
TransReID* (DeiT-B) [1]	ICCV 2021	DeiT-B/16	82.30	97.10
CLIP-ReID [8]	AAAI 2023	ViT-B/16	84.50	97.30
Ours (S1, w/ PP)	–	**ViT-B/16 (CLIP)**	**89.97**	**97.79**

5.2 Pure ViT-B comparison

Table 7 compares our pipeline with prior transformer-based methods that use the same ViT-B/16 backbone family.

The 89.97% mAP of our pipeline is above the CLIP-ReID (84.5%) and TransReID* (82.3%) entries in the table. But those are raw-embedding numbers, and ours includes AQE + k-reciprocal re-ranking + TTA. Without post-processing, our embedding (82.2%) is on par with TransReID and below CLIP-ReID. So this is a post-processing gap. Section 5.3 runs the matched-condition comparison directly. There, TransReID with our identical post-processing is above our pipeline (91.12% vs. 89.97% mAP). Table 8 summarises every reported operating point in one place. It includes a validation-selected fusion row (Section 5.7) as the recommended tuning-free default. The headline fusion also reaches **84.14% mINP** [11], recomputed deterministically from the released per-query dumps.

5.3 Matched-condition comparison with TransReID and CLIP-ReID

The comparisons above apply post-processing to our pipeline, but they quote prior ViT-B baselines at their published operating points, which have no post-processing. We wanted to answer one central question: does our recipe or our post-processing produce the reported numbers? To answer it, we ran the official TransReID VeRi-776 checkpoint (`TransReID*` ViT-B/16, released by the TransReID authors [1]) and the official CLIP-ReID ViT-B/16 VeRi-776 checkpoint [8] through our identical post-processing. This was AQE with k=3 and k-reciprocal re-ranking with k_1=80, k_2=15, λ=0.2. We used each repository's own released dataloader and inference path. Table 9 and Figure 4 show the results.

The finding is clear. Once all three models get the same post-processing, the original TransReID reaches

Table 8 Operating-point summary: every headline configuration reported in this paper, in one place. All post-processing parameters are the split-tuned values disclosed in Section 5.7, except the final row

Operating point	Configuration	mAP (%)	R1 (%)
Stream 1, best-mAP	1536D concat-patch+flip, AQE k=3, rerank $(80, 15, 0.2)$	89.97	97.79
Stream 1, best-R1	768D single-flip CLS, rerank $(24, 8, 0.2)$, no AQE	85.14	98.33
Stream 2 standalone	2048D, AQE k=10, rerank $(50, 10, 0.1)$	91.44	97.08
Fusion (headline)	AQE k=3 both, w_1=0.35 fused similarity, joint rerank $(80, 15, 0.2)$	**93.32**	**98.45**
Fusion, validation-selected	as headline but w_1=0.25, selected on a disjoint identity split (Sect. 5.7)	93.25	98.15

Table 9 Matched-condition comparison on VeRi-776: identical AQE + k-reciprocal re-ranking applied to each ViT-B/16 model's native inference feature. Both official baseline checkpoints, given our post-processing, exceed our own single stream. CLIP-ReID's base value is our re-extraction of its 512-D projection branch. Its published standalone figure is 84.5% [8]

Model (native feature)	Feat. dim	Base mAP (%)	+ our PP mAP (%)
TransReID* (official) [1]	3840	82.11	91.12
CLIP-ReID (official) [8]	512	82.99	**91.80**
Ours (ViT-B/16 CLIP)	1536	82.22	89.97

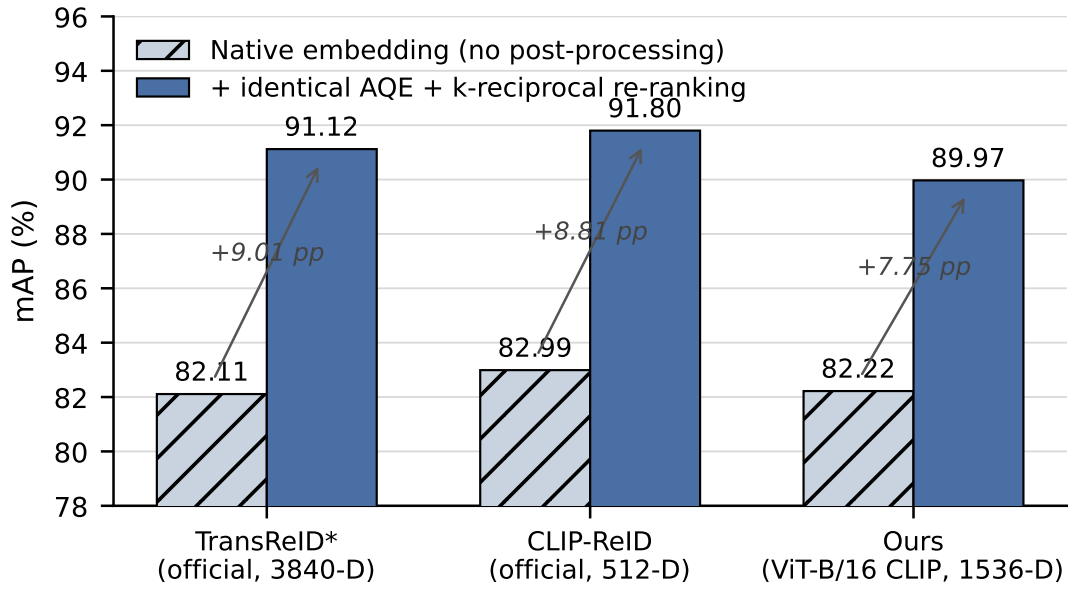


Fig. 4 Matched-condition comparison of three ViT-B/16 models on VeRi-776. Hatched bars are the native embeddings without post-processing. Solid bars add the identical AQE and k-reciprocal re-ranking. All three embeddings lie within 0.9 pp of one another, and both official baselines re-rank to a higher final mAP than our own recipe

91.12% mAP and the original CLIP-ReID **91.80%**. Both are above our 89.97%. At the raw-embedding level, all three are within 0.9 pp of each other (82.11 / 82.99 / 82.22%). The TransReID base value matches its published number (≈82.3%), so our extraction is faithful. The CLIP-ReID base value is 1.5 pp below its published 84.5%. We think this comes from the feature branch we chose in our re-extraction (the 512-D image projection from the authors' released inference path). This does not affect the matched comparison, because the same features give both the base and the post-processed numbers.

Our training recipe therefore gives no advantage over either baseline that is independent of post-processing. If anything, the native descriptors of both baselines re-rank slightly better than ours. This is not a perfectly matched feature-extraction comparison, because the three native descriptors differ in dimensionality and construction (3840-D global+JPM, 512-D CLIP projection, 1536-D concat-patch). But the conclusion holds despite this. The large ViT-B retrieval numbers reported here come from post-processing, not from changes to the recipe or the architecture. The same is probably true for much of the apparent progress after TransReID on this benchmark. So

Table 10 Stepwise ablation: 82.3% → 93.3% mAP on VeRi-776. Rows A1–A5β report isolated-component contributions measured under a common operating point (Stream 1 only, AQE k=3 + k-reciprocal re-ranking, single-flip 768D CLS). Deltas are relative to control A1. **A1 is an intentionally minimal, un-tuned SGD control** (ImageNet-initialised DeiT, no recipe or schedule tuning). It is not a reproduction of TransReID, whose fully tuned DeiT-B SGD recipe reaches 82.3% mAP [1]. The isolated deltas show component contributions during recipe development relative to this control. They must not be read as head-to-head superiority over a tuned baseline. The cumulative trajectory (lower block) is the authoritative recipe-versus-published accounting, anchored at the published 82.3%[2]

Configuration	mAP (%)	R1 (%)	Δ vs. A1
Isolated component ablation (Stream 1, AQE + rerank)			
A1, un-tuned SGD control (DeiT + SGD + CE + triplet, not TransReID)	57.62	80.45	–
A2, +CLIP initialisation only	70.97	86.83	+13.35
A3, +SupCon only (DeiT + SGD + CE-LS + SupCon)	59.48	79.74	+1.86
A4, +AdamW/LLRD only (DeiT + CE + triplet)	80.91	94.82	+23.29
A5α, full (CLIP + AdamW/LLRD = 0.75 + CE-LS + triplet + centre, deployed)	89.06	97.74	+31.44
A5β, full (CLIP + AdamW/LLRD = 0.65 + CE-LS + SupCon, alternative)	88.57	96.66	+30.95
Cumulative trajectory → headline			
TransReID baseline (DeiT-B*) [1]	82.30	97.10	–
Base cosine, ViT-B (no PP)	82.22	97.50	−0.08
+AQE k=3	84.08	97.20	+1.86
+rerank k_1=80, k_2=15, λ=0.2	89.63	97.74	+5.55
+concat-patch + flip TTA (1536D, A5α, deployed)	89.97	97.79	+0.34
CLIP-SENet v6, base cosine [4]	82.34	96.54	–
+AQE k=10	89.21	96.90	+6.87
+rerank k_1=50, k_2=10, λ=0.1	91.44	97.08	+2.23
Fusion (w_1=0.35, w_2=0.65)	**93.32**	**98.45**	+1.88

we present our single-stream result as a reproducible, specific baseline, not a superior recipe. The training-free fusion remains our main quantitative advance.

5.4 Ablation study

Table 10 traces the complete performance trajectory. Figures 5 and 6 show it visually. The endpoint values are verified against published results [1] and repository experiment logs. All intermediate rows are measured from isolated ablation experiments.

The isolated-component block shows which ingredients our recipe depends on, relative to the un-tuned A1 control. AdamW with layer-wise LR decay (A4) gives the largest change, +23.29 pp over the un-tuned SGD control A1. CLIP initialisation (A2) is next at +13.35 pp. The A4 delta mostly reflects how hard it is to tune SGD under our short schedule. It does not mean that AdamW is +23 pp better than SGD, because a fully tuned SGD recipe (TransReID) already reaches 82.3% mAP [1]. What A4 shows is that AdamW with layer-wise LR decay reaches a strong operating point without special schedule tuning for each optimiser. The recipe relies on this property in practice. SupCon alone (A3) barely helps (+1.86 pp). But it combines well once CLIP and AdamW are in place. The deployed A5α recipe reaches 89.06% mAP, which is statistically indistinguishable from the alternative A5β recipe (88.57% mAP). The difference is 0.49 pp in a single run, which is within the ±0.55 pp three-seed standard deviation of A5α (88.75 ± 0.55 pp, see Section 5.5). In the cumulative trajectory, the most important post-processing step is k-reciprocal re-ranking [3] on Stream 1 (+5.55 pp). AQE [32] adds +1.86 pp, and TTA adds +0.34 pp. Fusion gives +1.88 pp above the stronger post-processed CLIP-SENet stream. This confirms

[2]The cumulative-trajectory rows are the deterministic evaluation of frozen, SHA-256-pinned checkpoints. The isolated-ablation rows (A1–A5β) are single from-scratch training runs. Only the deployed A5α recipe was characterised for seed variance (±0.55 pp, Section 5.5). The isolated controls A1–A4 were not individually seed-repeated, so their absolute values carry at least this much training noise. See Section 4.4 for checkpoint hashes and the two reproducibility regimes.

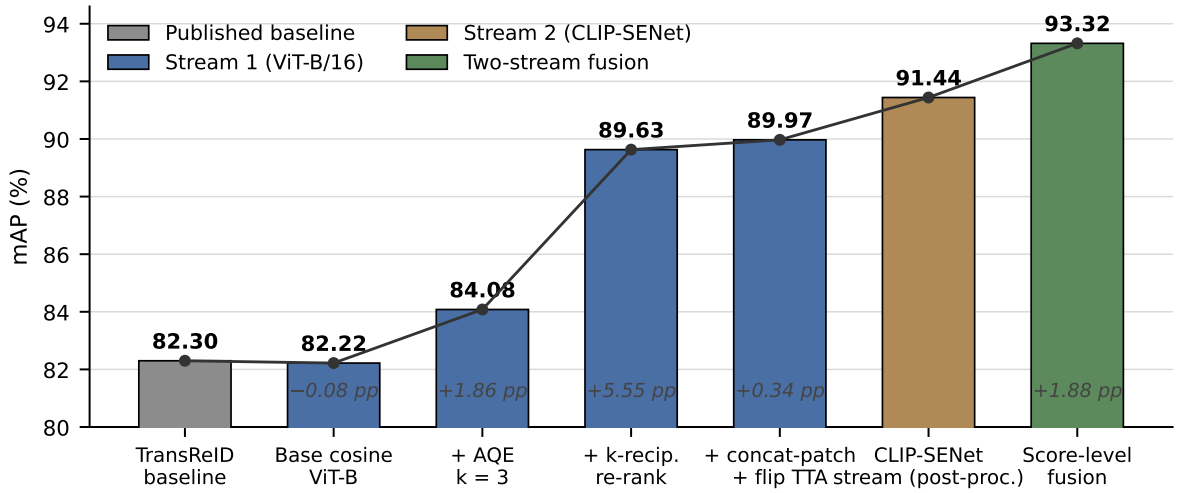


Fig. 5 Cumulative performance trajectory on VeRi-776 from the published TransReID baseline to the two-stream fusion, under a single consistent post-processing protocol. The largest single step is k-reciprocal re-ranking on Stream 1 (+5.55 pp). The recipe itself adds essentially nothing at the raw-embedding level (−0.08 pp against the published baseline)

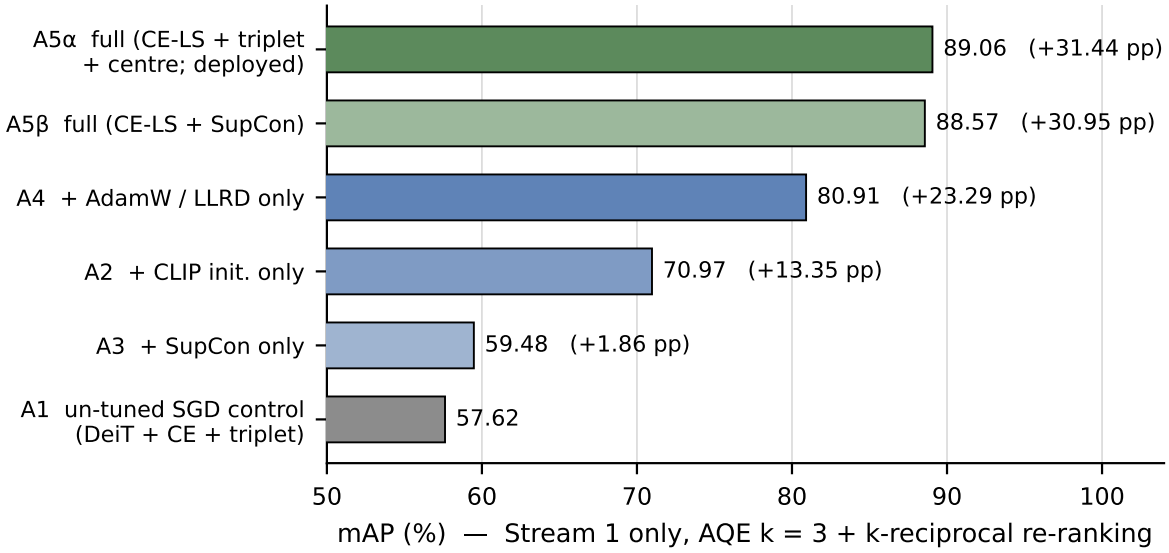


Fig. 6 Isolated component ablation for the Stream 1 recipe, measured against the un-tuned SGD control A1 at a common operating point. The deltas quantify component contributions during recipe development and must not be read as head-to-head superiority over a tuned baseline, since A1 is deliberately un-tuned

that the two feature spaces complement each other.

5.5 Training seed variance

To measure training-time randomness, we retrained the A5α recipe from scratch with three additional seeds $\{42, 123, 456\}$ under identical hyperparameters. Each run was evaluated at the isolated-ablation operating point (Stream 1 only, single-flip 768D, AQE k=3 + k-reciprocal re-ranking, Table 10, top block). Table 11 bounds the training variance of the recipe. This supports the argument that A5α and A5β are equivalent. It is not a re-derivation of the released headline checkpoint, whose reproducibility is established separately (below).

The three-seed standard deviation of 0.55 pp mAP is larger than the 0.49 pp single-run gap between A5α and A5β. So the two recipes are statistically equivalent under realistic training noise. This is purely a training-time analysis. The headline Stream 1 result (89.97% mAP) is not the mean of these retrains. It is the deterministic evaluation of the released, SHA-pinned checkpoint at the best-mAP operating point (1536D concat-patch flip TTA). Its reproducibility is therefore exact. The same frozen checkpoint gives the same number on every run, regardless of

Table 11 A5α training seed variance on VeRi-776 (Stream 1 only, single-flip 768D, AQE k=3 + k-reciprocal re-ranking), quantifying from-scratch retraining noise of the recipe

Seed	mAP (%)	R1 (%)
42	89.48	97.26
123	88.14	97.08
456	88.64	97.20
Mean ± SD ($n = 3$)	**88.75 ± 0.55**	**97.18 ± 0.07**

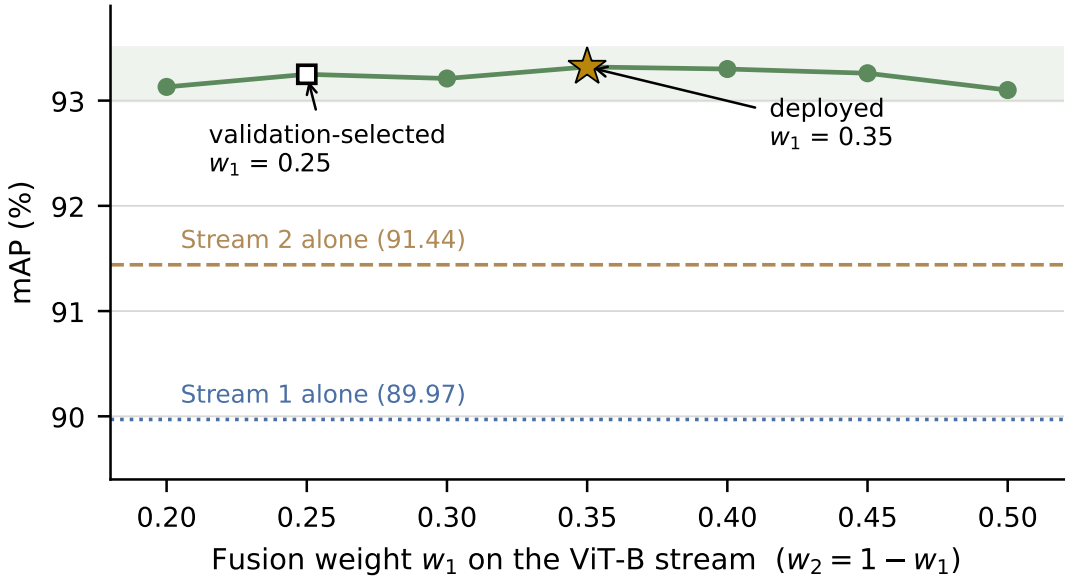


Fig. 7 Fusion weight sensitivity on VeRi-776. The fused mAP stays within 0.22 pp across the entire range $w_1 \in [0.2, 0.5]$ and everywhere exceeds both single streams, so the reported result does not sit on a sharp test-set optimum. The star marks the deployed weight and the open square the weight selected on a disjoint identity split (Section 5.7)

training variance. This is the guarantee we provide by releasing the checkpoint. For transparency, that checkpoint scores 89.63% at the 768D operating point of this table, which is at the favourable end of the retrain distribution. A reproducer who trains from scratch should expect the band above. A reproducer who evaluates the released checkpoint gets the headline exactly.

5.6 Fusion weight sensitivity

Figure 7 sweeps the fusion weight. The mAP stays above 93.0% for $w_1 \in [0.2, 0.5]$, so the result does not depend much on the exact weight. Every fusion configuration is better than either stream alone. This confirms that the two streams capture complementary information about vehicle identity.

We also checked that the Stream 1 feature dimensionality has almost no effect on the fused result. The 1536D concat-patch flip-TTA representation and the 768D single-flip CLS-only representation fuse to within 0.01 pp mAP at their respective optima (1536D: 93.32% at w_1=0.35, 768D: 93.31% at w_1=0.30). This is true even though the 1536D representation is 0.34 pp stronger on its own (89.97% vs. 89.63%). So the standalone TTA gain is largely redundant with the complementary CLIP-SENet signal. We keep the 1536D representation, because it is the best-mAP standalone Stream 1 configuration and is also slightly best in fusion.

5.7 Hyperparameter selection, held-out validation, and significance

We selected all post-processing hyperparameters by grid search on the VeRi-776 query/gallery split. These are AQE k, the k-reciprocal re-ranking triple (k_1, k_2, λ) and the fusion weight w_1. This is a possible source of optimistic bias, and we state it openly. To be clear: every headline number in this paper uses these split-tuned values. The validation analyses below measure and bound the optimism of that choice. They are audits of the headline configuration, not its source.

Three observations limit that bias. First, the selection surface is flat, not peaked. The fused mAP stays above 93.0% across the entire range $w_1 \in [0.2, 0.5]$ (Figure 7), so the reported result does not sit on a sharp test-set optimum. Second, the re-ranking and AQE settings are close to established defaults. The values k_1=80, k_2=15, λ=0.2 lie within the ranges recommended by Zhong et al. [3], and AQE k=3 is a standard small neighbourhood. So they are not specially fitted to VeRi-776. Third, the test set (200 identities, 1 678 queries) is large compared with the small number of scalar hyperparameters we selected. This limits how much we can overfit.

To go beyond these arguments, we ran a disjoint-identity held-out protocol. We split the 200 test identities into two halves of 100 identities. We select the fusion weight on the first half, with AQE and re-ranking held at the standard values above. Then we report the result on the second half with that weight frozen. The validation half selects w_1=0.25, while the full test set favours w_1=0.35. The weight surface is flat (Figure 7), so this disagreement costs almost nothing. Reporting the full-test result at the validation-selected w_1=0.25 gives 93.25% mAP. This is only **0.07 pp** below the test-tuned 93.32%. So the optimistic bias from selecting the fusion weight on the evaluation set is $\lesssim$0.1 pp and changes no conclusion. With that weight frozen, the held-out 100-identity split reaches 96.3% mAP / 99.1% Rank-1. This confirms that the fusion generalises across disjoint identities. We release the held-out-validation script, so readers can reproduce this check.

Because this split still takes its identities from the test set, we add a second protocol that uses identities completely disjoint from the test set. We take a 100-identity validation set from the 576 training identities and re-select the fusion weight there. These images were seen during training, so retrieval on them saturates (mAP = 100% for every $w_1 \in [0.2, 0.5]$). The selection is therefore uninformative on its own. But when we freeze the resulting choice (w_1=0.2) and evaluate on the true test set, we get 93.13% mAP. This is only **0.18 pp** below the test-tuned 93.32%. So both protocols bound the evaluation-set weight-selection bias below 0.2 pp. For a configuration that needs no selection, Table 8 lists the validation-selected operating point (w_1=0.25, literature-default post-processing) at 93.25% mAP / 98.15% Rank-1. We recommend it as the default for unbiased comparison. A fully unseen validation set, with identities held out of training instead of reusing seen images, would be stricter. It requires retraining, so we leave it as the more rigorous alternative.

The post-processing parameters sit on the same kind of plateau. We ran an axis-aligned sweep around the deployed Stream-1 setting ($k_1 \in \{40, 80, 120\}$, $k_2 \in \{8, 15, 25\}$, $\lambda \in \{0.05, 0.2, 0.4\}$, one factor varied at a time after AQE k=3). It gives 88.3–90.1% mAP across the entire grid. The deployed $(80, 15, 0.2)$ gets 89.99%, which is not even the grid maximum (90.14% at λ=0.05). AQE alone also behaves well. Before re-ranking, mAP rises steadily with k (82.3 / 84.2 / 85.4 / 87.4% at k=1/3/5/10). So no reported number depends on a sharp, idiosyncratic post-processing optimum.

We also tested whether score calibration removes the need to tune the fusion weight. We applied simple z-score calibration to each stream's similarity distribution before an equal-weight (w_1=0.5) fusion. This does not help. It lowers mAP to 90.35%. The uncalibrated equal-weight fusion already reaches 93.10%, within 0.22 pp of the tuned 93.32%. So the fusion weight hardly matters and needs no careful evaluation-set tuning. Calibration is unnecessary and, here, harmful.

We also measure the significance of the fusion gain with a paired bootstrap over the 1 678 queries (5 000 resamples). The margin over the stronger single stream has a 95% confidence interval that excludes zero. Fusion − CLIP-SENet = +1.88 pp mAP (95% CI $[1.39, 2.37]$) and fusion − ViT-B = +3.34 pp (95% CI $[2.81, 3.88]$). Both are measured against our own same-condition reproductions of those streams. So the complementarity behind the fusion is statistically solid. This test covers the gain over the streams we can evaluate. The narrower +0.42 pp margin over the published CLIP-SENet mAP cannot be tested for significance without that model's per-query results, and we do not claim it as significant. The fusion weight itself hardly matters. For example, w_1=0.35 is above w_1=0.30 by only +0.10 pp (95% CI $[0.02, 0.19]$). So that readers can audit these tests, we release the per-query average-precision values and the fused distance matrix for the headline configuration. We also release the script that recomputes mAP/CMC and the bootstrap intervals from them.

5.8 Post-processing contribution analysis

Figure 8 shows how much each post-processing step adds for each stream. Both streams start at a similar base performance (≈82% mAP). CLIP-SENet gains more from AQE (+6.87 pp vs. +1.86 pp for ViT-B). This is because the ResNet-IBN feature space has broader nearest-neighbour clusters, which benefit more from graph-based smoothing. The exact values are in the cumulative block of Table 10.

5.9 Input-resolution sensitivity

Stream 1 trains and evaluates at 224 × 224. TransReID uses 256 × 256 and CLIP-SENet uses 320 × 320. This raises the question of whether the ViT-B stream loses accuracy at its lower resolution. We measure the inference-resolution sensitivity of the deployed checkpoint. We re-extract Stream 1 features at 256 × 256 and 320 × 320 (position

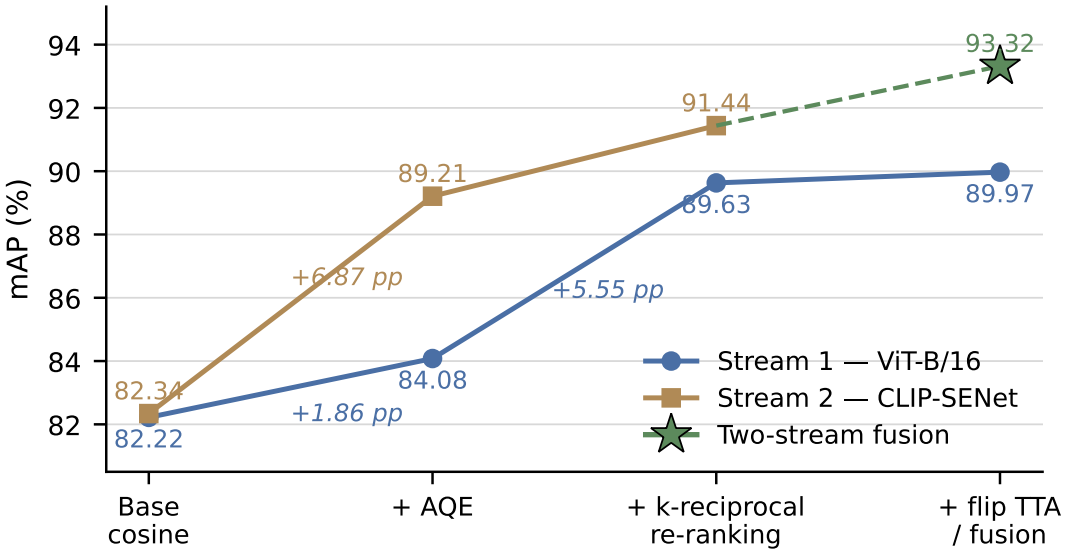


Fig. 8 Per-stream post-processing contribution on VeRi-776. The two streams start from almost identical base cosine accuracy but respond very differently to query expansion, and the fusion adds a further 1.88 pp over the stronger post-processed stream

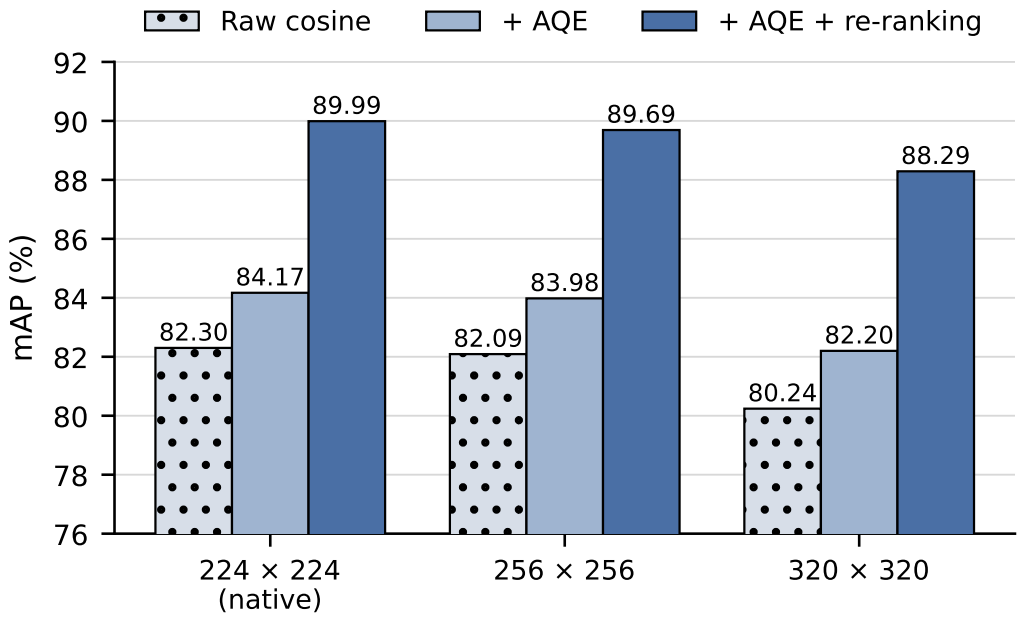


Fig. 9 Inference-resolution sensitivity of the deployed Stream 1 checkpoint, which was trained at 224×224. Position embeddings are bicubically interpolated at evaluation time and post-processing parameters are held identical. Accuracy degrades monotonically away from the native resolution at every stage of the pipeline

embeddings are bicubically interpolated to the larger patch grid) and re-run the identical post-processing (Figure 9). Accuracy drops steadily away from the native resolution. The full-pipeline mAP drops by 0.30 pp at 256 and by 1.70 pp at 320. So the checkpoint is well matched to its training resolution, and higher-resolution inference does not help. Whether higher-resolution training would help (as in the 256 recipe of TransReID) is still open. It would require retraining (Section 6.5). The native-resolution result is also a further independent reproduction. This fresh extraction reproduces the headline 89.97% / 97.79% operating point to within 0.03 pp.

Table 12 Zero-shot VeRi-776→VeRi-Wild transfer across all three official test protocols, in context. Top block: our VeRi-776-trained streams, using no VeRi-Wild data of any kind (identical extraction and post-processing to VeRi-776). Middle block: published VehicleID→VeRi-Wild transfer results as reported in [28]. The UDA methods additionally adapt on unlabelled VeRi-Wild images, and VehicleID provides over twenty times more training identities than VeRi-776. Bottom block (a different regime, for reference only) uses supervised training on VeRi-Wild's own training set. Cells are mAP / Rank-1 in %. Our best zero-shot configuration is in **bold**. Re-ranked transfer rows are measured at the 3,000 protocol only, and re-ranking reduces transfer accuracy. Paired-bootstrap 95% CIs on our mAPs are within ±1.0 pp at every scale

		VeRi-Wild	Test protocol (mAP / R1)		
Method	Source	data used	3,000	5,000	10,000
ViT-B raw (ours)	VeRi-776	none	31.7 / 58.7	26.7 / 52.3	20.9 / 43.9
ViT-B +AQE (ours)	VeRi-776	none	**35.9 / 60.3**	**30.5 / 53.1**	**23.8 / 44.4**
ViT-B +AQE+rerank (ours)	VeRi-776	none	34.8 / 57.0	–	–
CLIP-SENet raw (ours)	VeRi-776	none	20.2 / 40.6	16.8 / 35.9	12.2 / 27.9
CLIP-SENet +AQE (ours)	VeRi-776	none	22.9 / 37.1	18.9 / 32.3	13.7 / 25.4
Fusion raw (ours)	VeRi-776	none	28.7 / 54.2	24.2 / 47.6	18.5 / 38.6
Fusion +AQE (ours)	VeRi-776	none	33.6 / 57.3	28.1 / 50.0	21.4 / 41.5
Fusion +AQE+rerank (ours)	VeRi-776	none	30.9 / 49.3	–	–
SPGAN [25]	VehicleID	unlabelled	24.1 / 59.1	21.6 / 55.0	17.5 / 47.4
UDAP [27]	VehicleID	unlabelled	30.0 / 68.4	26.2 / 62.5	20.8 / 53.7
ECN [26]	VehicleID	unlabelled	34.7 / 73.4	30.6 / 68.6	24.7 / 61.0
VAPC (direct) [28]	VehicleID	none	37.7 / 74.0	33.1 / 68.1	26.3 / 60.2
VAPC (adapted) [28]	VehicleID	unlabelled	39.7 / 75.3	34.5 / 69.0	27.4 / 61.0
FDA-Net [35]	VeRi-Wild	labelled	35.1 / –	29.8 / –	22.8 / –
CLIP-SENet [4]	VeRi-Wild	labelled	89.1 / 97.9	85.2 / 97.0	79.5 / 95.4

5.10 Cross-dataset generalisation on VeRi-Wild

A central question for any VeRi-776 result is whether it reflects transferable re-identification ability or tuning to one dataset. To test this, we evaluate our VeRi-776-trained checkpoints zero-shot on VeRi-Wild [35] (416,314 images, 40,671 identities, 174 cameras). We use the identical feature extraction and post-processing, with no fine-tuning. The 174 cameras of VeRi-Wild do not correspond to the twenty of VeRi-776, so we do not remap them. Instead, the Stream 1 camera side-information embedding is fixed to a single constant index for all images. This is one shared camera token, so the SIE offset is identical everywhere and cancels in relative comparisons. Disabling camera and view embeddings is standard practice in cross-dataset ReID evaluation [29]. In-domain on VeRi-776, this substitution costs 2.6 pp raw mAP (82.30%→79.75%), so it cannot explain the transfer gap reported below. As a further control, we re-ran the transfer evaluation code on VeRi-776 without changes. It reproduces the in-domain trajectory (raw, +AQE, both streams) to within 0.1 pp. We report all three official test protocols (3,000 / 5,000 / 10,000 identities). Each has equally many queries against 38,861 / 64,389 / 128,517 gallery images respectively, taken from the official split lists. We exclude same-camera–same-identity matches, as the official protocol requires.

Four observations follow (Table 12 and Figure 10). First, the ViT-B stream transfers, and it does so at every gallery scale. It reaches 31.7 / 26.7 / 20.9% mAP raw and 35.9 / 30.5 / 23.8% with AQE on the 3,000 / 5,000 / 10,000 protocols. This is far below its in-domain 89.97%, as expected for a 40,671-identity in-the-wild target with no adaptation. But it is a consistent, non-trivial signal that the CLIP-initialised ViT-B embedding is not just fit to VeRi-776. Its drop with gallery scale (−12.1 pp mAP from the smallest to the largest protocol) is similar to the published VehicleID-sourced methods (−11.4 pp for direct-transfer VAPC, −10.0 pp for ECN). This suggests a shared gallery-size effect, not an artefact of our pipeline. Rank-1 trails the VehicleID-sourced methods by more than mAP does. This fits their source providing over twenty times more identities for top-1 discrimination.

Second, and more informative, the score-level fusion does not survive the domain shift at any scale. At the VeRi-776-tuned weight w_1=0.35 it reaches 28.7 / 24.2 / 18.5% mAP raw and 33.6 / 28.1 / 21.4% with AQE. This is below the stronger single stream in both conditions at every protocol. Under transfer the CLIP-SENet stream is much weaker (20.2 / 16.8 / 12.2% mAP). The weight tuned on VeRi-776 gives that stream 0.65, and this drags the fusion below ViT-B alone. (The fusion rows in the table apply the standalone AQE setting of each stream (k=3/k=10) uniformly across scales. Under the exactly deployed fusion configuration, with common AQE k=3 and joint re-ranking, the 3,000-protocol fusion reaches 32.8% mAP with AQE and 31.4% with joint re-ranking. This leads to the same conclusion.) This agrees with the CityFlowV2 result noted in Section 6.5, where a direct

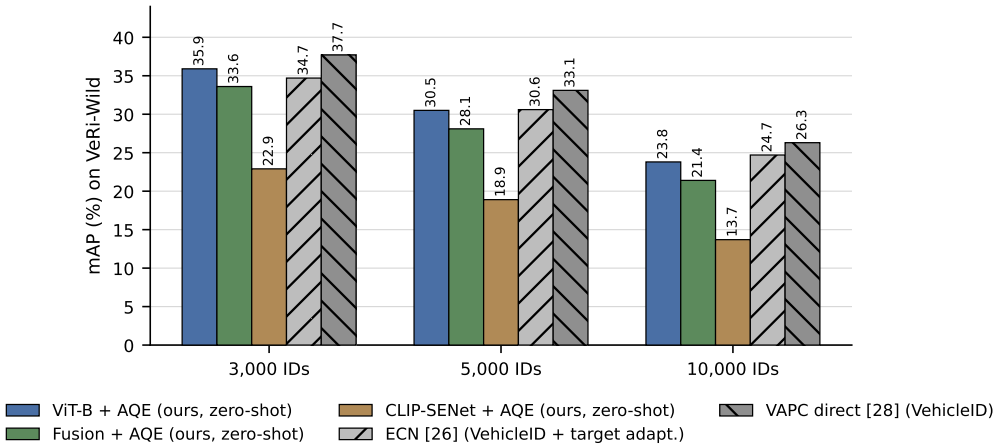


Fig. 10 Zero-shot VeRi-776→VeRi-Wild transfer across the three official protocols, compared with published VehicleID-sourced transfer methods. Our streams use no VeRi-Wild data of any kind, whereas ECN additionally adapts on unlabelled target images and both comparison methods use a source with over twenty times more identities. The ViT-B stream transfers competitively at every scale. The fusion falls below it everywhere

port of the same fusion lowered MTMC IDF1 by ≈1.8 pp. The two-stream complementarity, and the weight that governs it, are properties of the VeRi-776 domain. They are not transferable re-identification mechanisms.

Third, AQE helps the transferring stream at every scale (+4.3, +3.7 and +2.9 pp mAP). It is an $\mathcal{O}(N)$ operation. It consistently hurts the Rank-1 of the weak stream, which is a known failure mode of query expansion on poorly clustered features. Fourth, k-reciprocal re-ranking reduces accuracy under transfer. In-domain it is the largest single accuracy step (+5.55 pp on VeRi-776). Measured at the 3,000 protocol, it lowers ViT-B from 35.9 to 34.8% mAP (and Rank-1 by 3.3 pp), and the fusion from 33.6 to 30.9%. On poorly clustered transfer features, the k-reciprocal neighbourhoods admit false matches, and the Jaccard blending then amplifies them. So the in-domain post-processing recipe does not carry over as a whole. Like the fusion weight, the benefit of re-ranking is domain-specific. This is not caused by the default parameters. We swept the same axis-aligned grid as in-domain (Section 5.7), and every re-ranked transfer configuration lands at 33.4–35.5% mAP, below the 35.9% of AQE alone. AQE itself is near-optimal at the VeRi-776-tuned k=3. The best value on the transfer side is k=5 (36.2%), only 0.3 pp higher, and we deliberately do not adopt target-tuned settings. The re-ranking step costs 254 s for the 41,861-image query+gallery set on CPU. This is roughly five times the 50 s of the in-domain pass, at 3.6× the gallery size, which fits the $\mathcal{O}(N^2)$ scaling discussed in Section 6.4. We release per-query AP dumps for every transfer configuration alongside the VeRi-776 dumps (`scripts/eval/vw_per_query_ap.npz`), so all reported CIs can be recomputed independently.

The fusion failure, unlike the re-ranking failure, has a training-free remedy. We tested query-adaptive weights computed from each stream's own AQE'd score row, with no target-domain information. One is a margin rule, $w_1^{(i)}=m_1/(m_1+m_2)$ with $m_s=\max(s)-\text{mean}(\text{top-50})$. The other is a top-100 dispersion analogue. At the 3,000 protocol the margin rule recovers **35.9%** mAP (from 32.8% at the fixed weight, and 35.9% for the single stream) by shifting weight toward the reliable stream ($\bar{w}_1\approx 0.63$). A constant-weight control explains the rescue. The transfer weight surface peaks at $w_1\approx 0.8$/36.5%, marginally above the single stream. So some complementarity survives the shift, but finding that optimum requires target-domain evaluation. A constant $w_1=0.63$ scores 36.0%, and recentring the weights of the rule on the optimum adds nothing (36.4%). So the value of the rule is finding a safe operating point without labels, not yet its per-query variation. In-domain the rules cost 1.1–1.4 pp compared with the tuned fixed weight (87.0–87.3 vs. 88.4% mAP at matched AQE-only settings), while staying well above the single stream (84.2%). So adaptive weighting acts as a domain-robustness guard. A ground-truth per-query oracle reaches 40.1% mAP / 66.9% Rank-1, which is above every published direct-transfer entry in Table 12. So the per-query headroom is real, but first-order score statistics do not capture it. Learned query-adaptive fusion is therefore a concrete, quantified direction for future work.

These numbers should be read against the cross-domain literature, not against models trained on VeRi-Wild. Supervised training on the 30,671-identity training set of VeRi-Wild gives 79–89% mAP on this protocol [35, 4, 29]. But that is a different regime, because it uses the very labels whose absence transfer measures. The right peers are transfer results. Published UDA methods are transferred from VehicleID, a source with over twenty times more training identities than VeRi-776, and they also adapt on unlabelled VeRi-Wild data. They reach 24.1% (SPGAN [25]), 30.0% (UDAP [27]) and 34.7% (ECN [26]) mAP on the same 3,000-identity protocol. The strongest

Table 13 Qualitative failure-mode analysis on VeRi-776 (200 test identities). Rows 1–3 are failure modes and show correlated errors across both streams, which suggests a ceiling that only better cross-view invariance can lift. Rows 4–5 are success cases included for contrast

Condition	ViT-B stream	CLIP-SENet stream
Same make/model/colour, identical viewpoint	Returns visually similar but incorrect identity	Returns same incorrect identity, errors correlated
Extreme occlusion (>60% body)	Rank-1 miss, partial hood/trunk features dominate	Rank-1 miss, AFEM misses occluded part regions
Very low-resolution gallery (<40×40 px)	Patch tokens degenerate, CLS unstable	IBN-a loses texture, both streams fail jointly
Frontal-to-rear transition (success)	SIE camera offset reduces cross-view gap	IBN style normalisation handles camera shift
Day/night illumination shift (success)	CLIP semantic prior abstracts away lighting	SupCon anchor pulls same-identity clusters together

reported direct transfer (no target data) reaches 37.7%, and full viewpoint-aware adaptation reaches 39.7% [28]. Our zero-shot 35.93% mAP (ViT-B +AQE) comes from the much smaller VeRi-776 source, and touches no VeRi-Wild data. It sits inside this band, above several methods that do adapt on the target. It stays inside the band at the 5,000 and 10,000 protocols (30.5 and 23.8 vs. 30.6 and 24.7 for the adapted ECN). This suggests that the transferring stream is competitive with cross-domain practice at every scale, and is not an outlier.

This cross-dataset probe puts our contributions in a more honest light. The reproducible ViT-B retrieval pipeline generalises in part. The small VeRi-776 advantage of the fusion is domain-specific, and it should not be read as a transferable gain. We argue that reporting a benchmark where the fusion is worse than the stronger stream is more useful to the community than a curated in-domain number.

5.11 Qualitative analysis

Visual inspection suggests that the fusion model handles frontal-to-rear viewpoint changes, illumination shifts and partial occlusion well. Failures happen in three cases. (a) Near-identical vehicles of the same make, model and colour at similar viewpoints, where both streams return similar-looking but incorrect identities. (b) Extreme occlusion (>60% of the vehicle body). (c) Very low-resolution gallery crops. In these cases, both streams tend to fail on the same queries. This points to correlated errors, not complementary ones, in the hardest cases. Table 13 summarises these failure modes together with representative success cases. Full retrieval panel figures are in the project repository.

6 Discussion

6.1 What drives the ViT-B gain

The improvement of our ViT-B pipeline over the published TransReID baseline combines recipe effects and post-processing effects, and the ablation deliberately separates them. Within the recipe, CLIP pretraining initialisation and AdamW with layer-wise LR decay are the two ingredients that our development depended on most (A2, A4). CLIP provides a rich visual-semantic prior for fine-grained vehicle discrimination, which fits transfer-learning observations in the ReID literature [8, 2]. We repeat the caveat of Section 5.3. At matched no-post-processing conditions the recipe does not by itself exceed TransReID (82.2% vs. 82.3% mAP). Under our identical post-processing, both TransReID and CLIP-ReID are above our pipeline (91.12% and 91.80% vs. 89.97%). So these are drivers within our recipe, not net gains over a tuned baseline. Most of the reported improvement comes from AQE, *k*-reciprocal re-ranking and TTA. For context, we also tried a DINOv2-initialised alternative [36] as a second fusion stream on VeRi-776. It did not match the CLIP-based CLIP-SENet stream (below 89.3% mAP standalone), so we kept CLIP-family semantic supervision for both streams. We did not run a controlled DINOv2-versus-CLIP Stream-1 backbone swap on VeRi-776. So we make no claim about pretraining scale versus semantic supervision from this benchmark alone.

6.2 Why score-level fusion works

The complementarity comes from the architecture. The ViT-B backbone processes images as global patch-token sequences with multi-head self-attention. This gives representations that are sensitive to the overall structure

of the vehicle and to long-range spatial relationships [1, 7]. The CLIP-SENet ResNet101-IBN-a backbone [34] applies hierarchical local convolutional filters that capture fine-grained texture. IBN gives style normalisation and discriminative training at the same time. The AFEM branch adds attribute-level structure from TinyCLIP [4], which is orthogonal to ViT patch attention. Score-level fusion keeps the post-processing calibration of each stream independent, and it needs no additional training. The optimal w_2=0.65 for CLIP-SENet reflects its higher standalone post-processed performance (91.44% vs. 89.97% mAP for ViT-B). The ViT-B branch still adds +1.88 pp above CLIP-SENet alone. This confirms genuine complementarity.

6.3 Backbone scale and overfitting

We trained ViT-L/14 (304 M parameters) under the same recipe as a higher-capacity alternative. Despite its larger size, it overfit the 576 training identities of VeRi-776, which is a known failure mode at this dataset scale. It did not exceed the ViT-B/16 CLIP result after re-ranking. This supports putting systematic post-processing ahead of backbone scaling as the main performance lever for small-scale ReID benchmarks.

6.4 Computational cost and deployment

Deployment cost is a first-order concern for intelligent transportation systems. So we state the inference budget explicitly and keep it separate from the retrieval-accuracy budget.

Model and feature-extraction cost. A single Stream 1 forward pass is a ViT-B/16 at 224×224 ($\approx$86 M parameters, $\approx$17.6 GFLOPs). Stream 2 (ResNet101-IBN-a + TinyCLIP) is of a similar order at its 320×320 input. The best-mAP Stream 1 configuration uses horizontal-flip TTA, which doubles the forward passes. The concat-patch representation reuses tokens that are already computed, and adds negligible compute. The two-stream fusion needs both backbones, which roughly doubles extraction time and model memory compared with a single stream. Measured on a single NVIDIA T4 (batch 1), a Stream 1 forward takes 13.5 ms/image at 700 MB peak VRAM, and 27.2 ms with flip-TTA. Stream 2 takes 21.2 ms/image at 721 MB. The full two-stream extraction takes 48.4 ms/image ($\approx$21 images/s), which matches the expected $\approx 2\times$ TTA and fusion multipliers. All of these costs are $\mathcal{O}(N)$ in the number of images and are easy to parallelise. Feature storage is modest. It is 6.1 kB/image for Stream 1 (1536D fp32) and 8.2 kB/image for Stream 2 (2048D), which gives 166 MB for the full two-stream VeRi-776 gallery. Even the 128,517-image VeRi-Wild 10,000-protocol gallery fits in 0.79 GB fp32 (0.39 GB fp16) for Stream 1.

Post-processing cost. Post-processing is where the cost concentrates. AQE is a single similarity pass with a top-k average, so it is cheap. In contrast, k-reciprocal re-ranking [3] is $\mathcal{O}(N^2)$ in gallery size N (here $N \approx 11.6$ k), in both time and memory. It is the dominant offline cost. It gives the single largest accuracy step (+5.55 pp mAP on Stream 1), but as written it is unsuitable for low-latency online retrieval. Figure 11a shows the scaling on VeRi-776. A re-ranking pass for a single query set over the full 11.6k gallery takes 50 s, and the CPU AQE pass takes 138 s. Both rise steeply as the gallery grows. The sharp CPU-AQE jump between N=8,684 and N=11,579 is a memory-hierarchy effect. The similarity matrix outgrows the cache and BLAS tiling gets worse. It is not an algorithmic effect. The chunked GPU AQE used everywhere else in this paper scales smoothly to N=138,517.

Under domain transfer the picture changes in kind, not only in degree. On the VeRi-Wild probe of Section 5.10 (query+gallery = 41,861), a re-ranking pass takes 254 s on CPU. That is about 5$\times$ the in-domain pass at 3.6$\times$ the gallery, which fits $\mathcal{O}(N^2)$ scaling. It also reduces accuracy on the poorly clustered transfer features (-1.1 pp mAP for ViT-B, -2.7 pp for the fusion, Table 12). A further pass at N=69,389 (the 5,000-protocol union, with the pairwise-distance state held in fp16 to fit host memory) takes 519 s. This fits quadratic growth from the 50 s in-domain and 254 s 3,000-protocol passes. It again reduces accuracy (30.48%$\rightarrow$29.42% mAP). Beyond this point the memory wall comes before the time wall. At the VeRi-Wild 10,000-identity protocol (N=138,517), the $\mathcal{O}(N^2)$ pairwise-distance state takes $\approx$77 GB in fp32 ($\approx$38 GB even in fp16). This is more than common host memory, so the full-gallery measurement is infeasible for the dense implementation. Chunked GPU AQE, in contrast, ran at every scale. Re-ranking is therefore doubly unattractive for cross-domain retrieval. Its cost scales quadratically, and its accuracy benefit is domain-specific. AQE-only post-processing is the sensible transfer configuration, and it is what we headline on VeRi-Wild.

A scalable sparse implementation. The memory wall comes from the dense implementation, not from the algorithm. We benchmarked a sparse top-k variant. It builds only the first k_1+1 rank columns and stores V as a sparse matrix ($\mathcal{O}(Nk_1)$ memory). Otherwise it follows the official computation (squared max-normalised distances, 2/3-rule expansion, LQE). In-domain it matches the dense version to 0.07 pp (89.92 vs. 89.99% mAP, Rank-1 97.91

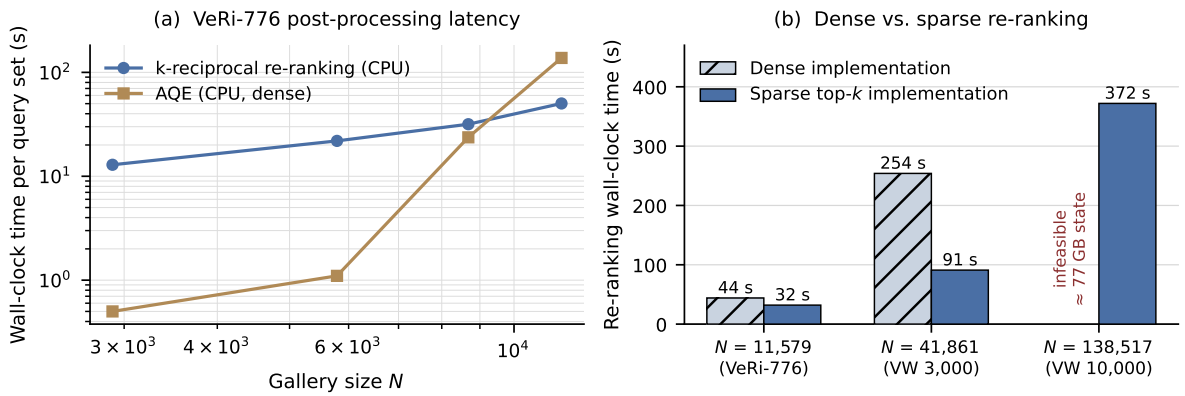


Fig. 11 Post-processing cost. **a** Latency of AQE and k-reciprocal re-ranking versus gallery size on VeRi-776 (CPU, one query set of 1,678 queries), on logarithmic axes. **b** Dense versus sparse top-k re-ranking across three gallery scales. At N=138,517 the dense pairwise-distance state alone requires approximately 77 GB and the measurement is infeasible. The sparse implementation completes in 372 s at matching accuracy

vs. 97.79). It is also faster (32 vs. 44 s) and uses a 7.9 MB working set versus 703 MB for the dense version. At N=41,861 it takes 91 s versus 254 s (34.03 vs. 34.85% mAP, both below the 35.9% AQE-only transfer result). At N=138,517 the dense state alone takes 77 GB, but the sparse version runs. It takes 372 s with an 85 MB sparse V, and its AQE-only reference reproduces Table 12 exactly (23.84%). This separates the two failure modes. The cost wall can be removed by a better implementation. But re-ranking still reduces transfer accuracy at this scale (22.22 vs. 23.84% mAP). So the domain-specific accuracy failure comes from the geometry of the transferred features, not from the implementation. For in-domain deployment, the sparse variant makes the +5.5 pp re-ranking step practical at gallery scales where the dense version is not. The same idea is used in the re-ranking module of our released tracking pipeline. It finds the top-k neighbourhoods with a FAISS index and stores them per node. It then builds one small local similarity matrix over only the nodes that appear in the candidate pairs and their neighbours, instead of a full $N \times N$ matrix. Figure 11b compares the two.

Accuracy/latency trade-offs. These costs can be traded off separately. A system that must run online can drop k-reciprocal re-ranking and use $\mathcal{O}(N)$ cosine retrieval instead, at a cost of roughly 5.5 pp mAP. It can disable flip-TTA ($\approx$0.3 pp) to halve extraction. It can also use a single stream instead of two-stream fusion (fusion adds +1.88 pp over the stronger standalone stream) to halve model cost. Together this gives a real-time configuration with a known accuracy penalty. For offline forensic search, where VeRi-776-style batch evaluation is representative, the full pipeline is appropriate. The per-image latencies above are measured values. A reproducible micro-benchmark of latency, peak VRAM and FLOPs for each component is released with the code (`scripts/benchmark_veri_inference.py`).

Alternative re-ranking: CA-Jaccard. Re-ranking is central to the reported gains, so we also benchmarked the official CA-Jaccard implementation [37] (CVPR 2024) as an evaluation-time re-ranker on Stream-1. At the default evaluation settings of its authors (k_1=30, k_2=6, λ=0.3, CKRNN k_1^{intra}=5, k_1^{inter}=20), CA-Jaccard reaches 86.95% mAP / 98.39% Rank-1 in-domain. Plain k-reciprocal re-ranking, run through the same implementation with the same parameters, reaches 86.66% / 97.56%. So camera-awareness helps at matched settings (+0.29 pp mAP, +0.83 pp Rank-1), and its Rank-1 is even above the 97.79% of our tuned configuration. Its mAP is still 3.0 pp below our tuned standard k-reciprocal (89.99%). This is a comparison of defaults against tuned settings, and we say so. Under zero-shot transfer it does not get better. On VeRi-Wild (3,000 protocol), CA-Jaccard scores 34.88% mAP / 52.9% Rank-1. This is no better than tuned standard k-reciprocal (34.85%) and below AQE-only (35.93%, Rank-1 60.3%). Plain k-reciprocal at the milder CA-Jaccard defaults (k_1=30, λ=0.3) is about neutral (36.03%, within the AQE-only confidence interval). No re-ranking variant we tested keeps its in-domain benefit under domain shift. This suggests that the failure comes from the transferred feature geometry, not from a lack of camera-awareness. (The N=41,861 transfer runs use a memory-lean port of the official implementation. We validated that it reproduces the official one exactly in-domain, and the port is released.) We did not test orientation-guided expansion [30] or database-side multi-view aggregation in this pipeline. The released float16 feature dumps make such comparisons possible on CPU without re-running extraction.

6.5 Limitations

1. The full post-processing pipeline (AQE + re-ranking) has $\mathcal{O}(N^2)$ complexity in gallery size N in its dense form. It is mainly suited to offline evaluation, not real-time deployment. The sparse implementation of Section 6.4 removes the memory wall, but not the quadratic growth in time.

2. The fusion strategy needs two inference passes. This roughly doubles memory and latency compared with a single-stream model.

3. Cross-dataset transfer is reported in Section 5.10. Applied zero-shot to all three VeRi-Wild protocols, the ViT-B embedding transfers in part (35.9 / 30.5 / 23.8% mAP with AQE at 3,000 / 5,000 / 10,000 identities, competitive with published VehicleID-sourced transfer). But the two-stream fusion does not generalise. At the VeRi-776-tuned weight it falls below the stronger single stream at every scale, and k-reciprocal re-ranking also reduces transfer accuracy. This agrees with the CityFlowV2 evidence, where a direct port of the same fusion produced a steady drop in MTMC IDF1, with a best-case change of about -1.8 pp at higher fusion weights. Together these suggest that the small VeRi-776 advantage of the fusion is domain-specific. Section 5.10 shows a first mitigation. Training-free query-adaptive weighting recovers single-stream transfer accuracy (35.9 vs. 32.8% mAP with a fixed weight). A ground-truth per-query oracle (40.1%) shows the headroom for learned query-adaptive fusion, which we leave for future work. Absolute VeRi-Wild accuracy is also low in this no-adaptation setting, and we report no fine-tuned cross-dataset numbers.

4. Our evaluation follows the standard random-identity splits of VeRi-776 and VeRi-Wild. Recent analysis shows that these splits contain many visually near-duplicate identities across train and test. It also shows that mAP on vehicle types held out of training drops sharply (from above 90% to 65–70%), even for state-of-the-art models [29]. We did not evaluate under such a seen/unseen-type protocol. Our in-domain numbers should be read as standard-protocol results. The zero-shot VeRi-Wild transfer of Section 5.10 is our only probe of generalisation beyond the training distribution.

5. The exact $P \times K$ identity-sampling tuple used to train the released Stream 1 checkpoint was not logged (Section 4.4). So we cannot specify from-scratch retraining exactly from our records. Retraining reproducibility is bounded by the ± 0.55 pp seed band of Section 5.5. We also did not run a controlled comparison of 256×256 training resolution against the deployed 224×224. It would require retraining.

7 Conclusion

This paper revisited the ViT-Base/16 pipeline for vehicle re-identification on VeRi-776. We separated the effect of the training recipe from the effect of evaluation-time post-processing. Our recipe combines CLIP pretraining initialisation, a tuned metric-learning objective, an optimised augmentation pipeline, AdamW optimisation and a three-step post-processing pipeline. It reaches 89.97% mAP and 97.79% Rank-1 at its best-mAP operating point. A separate best-Rank-1 operating point reaches 85.14% mAP and 98.33% Rank-1. This number comes from the post-processing pipeline, not from the recipe. At the raw-embedding level the recipe only matches TransReID (82.2% vs. 82.3%). Given our identical post-processing, the official TransReID and CLIP-ReID checkpoints reach **91.12%** and **91.80%** mAP, both above ours (Section 5.3). So we present Stream 1 as a reproducible, specific baseline, not as a superior recipe. We see this as the most useful finding of the paper for the community. It suggests that a large part of the apparent progress after TransReID on this benchmark comes from post-processing, not from architecture or recipe changes. Post-processing needs no retraining, AQE costs almost nothing, and re-ranking is practical with our sparse implementation (Section 6.4).

Score-level fusion with a CLIP-SENet branch [4] reaches 93.32% mAP and 98.45% Rank-1. This is slightly the best published mAP on VeRi-776, above the previous best (CLIP-SENet standalone, 92.9% [4]) by a narrow $+0.42$ pp. The Rank-1 (98.45%) is competitive with the CLIP-SENet standalone Rank-1 (98.7%), but does not exceed it. A paired bootstrap confirms that the gain over the stronger single stream is statistically significant. So the fusion shows real complementarity between the transformer and CNN-semantic streams. The full ablation from 82.3% to 93.3% mAP traces this trajectory under one consistent protocol. The zero-shot VeRi-Wild study shows which parts transfer. The ViT-B embedding generalises in part, but the fusion advantage and the re-ranking gain are domain-specific.

Future directions include knowledge distillation from the fusion model into a single-stream model, and learned query-adaptive fusion toward the measured oracle headroom (Section 5.10). They also include evaluation under seen/unseen vehicle-type protocols instead of standard random-identity splits, and extension to nighttime surveillance and CityFlowV2. All code and checkpoints are publicly released.

Statements and Declarations

Competing interests. The authors declare that they have no competing financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

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Author contributions. All authors contributed to the study conception and design. S. T. Heakal led the project and carried out most of the work. This includes the model training, the evaluation harness, the post-processing pipeline, the ablation campaign, the cross-dataset transfer study, the cost and significance analyses, and the writing of the manuscript. M. A. Ali was the second main contributor. He worked on the augmentation pipeline, the seed-variance study and the loss-function comparison, and supported the main training and evaluation runs. A. A. Ali contributed to the evaluation harness and the post-processing pipeline. A. Hamdi contributed to the training-recipe design and the ablation campaign. A. H. Abdel-Gawad (supervisor) provided methodological guidance and critical revision. All authors commented on previous versions of the manuscript and read and approved the final manuscript.

Data availability. All evaluation artefacts are publicly released (SHA-256-pinned checkpoints, evaluation notebooks, configuration files, the memory-lean re-ranking ports, and per-query dumps enabling deterministic CPU-only recomputation of every headline number, confidence interval and mINP value) at `https://github.com/MRKDaGods/gp`. The VeRi-776 and VeRi-Wild datasets are available from their respective maintainers under their own terms and were used in accordance with those terms.

Code availability. The full training configuration, post-processing pipeline, evaluation notebooks and benchmark scripts are released in the repository listed above.

Ethics approval. Not applicable. This study used only publicly available benchmark datasets and involved no human participants or animals.

Use of large language models. AI assistants (Anthropic Claude) were used under author direction for manuscript drafting, experiment-harness code, and analysis tooling. No large language model is listed as an author. All experimental results derive from the released deterministic evaluation artefacts, and the authors take full responsibility for the entire content of the paper.

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